On the Elasticity of Spacetime Curvature and Gravity as Its Felt Consequence: A Set of Working Hypotheses on Gravity, Space, and Time

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Abstract

General relativity identifies gravity with spacetime curvature: mass-energy curves spacetime, and free bodies follow the resulting geodesics. This paper sets out two working hypotheses that revisit that identification from opposite directions. Hypothesis 1 proposes that space itself is an elastic medium, and that the amount of curvature a given mass produces is governed by that medium's elasticity, so that the strength of gravity is tied directly to the elastic response of space rather than treated as a bare coupling constant. Hypothesis 2 makes that identification the paper's defining statement of gravity rather than a background assumption: spacetime curvature, sourced by mass, is the cause, and the felt force called gravity is its direct consequence – specifically, the widening gap between the local flow of time at neighboring points as curvature deepens between them. Two candidate drivers for the felt sensation of weight were checked against this definition: a radial, mass-centered one (curvature itself, carrying real, measurable tidal curvature – this paper calls it "centripetal" as a label for that radial, curvature- sourced term, not in the stricter textbook sense of the sideways acceleration that holds a body on a circular path; see §4.3 for the distinction) and a centrifugal one (a constant rotation of space). Only the radial, curvature-sourced case satisfies the definition; the centrifugal case reproduces the same felt sensation, via the equivalence principle, while carrying no true curvature at all, and is kept in this paper specifically to show what gravity is *not*. An earlier attempt to source the radial case from cosmic expansion's own acceleration was checked numerically and found roughly ten orders of magnitude too weak, confirming curvature itself, not any external driver, as the right place to locate the cause – though, as §4.3 and §7 make explicit, locating the cause this way is the Newtonian limit of the ordinary geodesic equation, not content beyond general relativity. Both hypotheses rest on a shared departure from the standard picture: treating space not as an empty, static backdrop but as a real

medium with its own fluidity and elasticity. Where the analogies used to develop that picture break against what is already established – several demonstrably do – this paper treats those breaks, in general, as marking where further observational and experimental work is needed on the medium's own properties, rather than as grounds to abandon the picture as a whole; where a break is instead a direct measurement against a specific claim's own stated prediction (as with H2a's original cosmic-expansion engine, H4's literal claim, and H6's scale claim; §7), that specific claim is withdrawn outright rather than merely flagged for later work – the distinction is between the research program and any one claim made under it. Both hypotheses are stated here in their roughest form, as working ideas to be formalized, checked against known limits, and tested against observational constraints – not as completed theories. Some of general relativity's quantitative successes are already recovered here, not merely promised: H1 reproduces inverse-square attraction, $g = GM/r^2$, by construction (§4.1), and H2c's deflection split – the standard GR formula, transcribed rather than independently derived from this paper's own machinery – is evaluated numerically against the measured solar light-bending angle of $1.75''$ and matches it; the same mechanism's perihelion-advance formula, evaluated for Mercury, reproduces the measured $42.98''$/century excess precession (§4.3). Gravitational-wave speed is not yet checked against this paper's framework; establishing it is the next item of future work flagged throughout.

A note on scope. H1 and H2 remain this paper's founding pair, but the shared foundation they rest on (§3) turned out to support several further, related hypotheses as the paper developed – currently H1 through H10, with H1b as a flagged extension of H1 – each carried to the same standard: an honest tier assessment (standard, extrapolated, or speculative; §2), and a numeric check wherever one could be run. Confidence varies sharply across them, by design: §2 and §7 state plainly which are kept as live, developed claims (H2a, revised – gravity as the felt consequence of spacetime curvature, this paper's defining statement, with the GPS correction, Chou et al.'s clock data, and Earth's tidal gradient all checked against it – though, as noted where it is derived, these are three numerical manifestations of the same single, already-standard potential $\Phi = -GM/r$ for Earth, not three independent confirmations of anything beyond it; H1, scoped; H7, without its extraction claim; H8; H9, as an adopted and already-bounded existing program) and which are better read as retired or substantially narrowed (H2a's original cosmic-expansion engine, specifically, superseded by its own revision rather than needed; H4's literal claim; H6's claim that its mechanism reaches the scale it was built to explain) – and which are kept only as a motivated, honestly-failing attempt at a still-open question (H10, §4.11: spacetime's elasticity as induced by the quantum vacuum rather than assumed, which currently misses the measured value by roughly thirty orders of magnitude) – so a reader is never left to guess which of the ten the paper is actually prepared to stand on.

1 Introduction

Summary. This paper starts from a single departure from the standard picture: treat space not as an inert backdrop that mass happens to curve, but as a real medium with its own elasticity and fluidity, properties to be measured rather than assumed away (H1, H1b). Under that picture, gravity itself is reframed as the felt consequence of curvature rather than an added force: the widening gap between the local rate of clock flow at neighboring points, as mass-sourced curvature between them deepens, *is* the acceleration

– a restatement of the Newtonian limit of general relativity's own geodesic equation, not a mechanism beyond it (H2a), with a spatial bend alone shown to be too weak, and a flat-spacetime rotational alternative shown to carry no true curvature, to do the job on their own (H2b, H2c). The same departure is then pressed against a cluster of standing astrophysical puzzles – what seeds the largest black holes, what seeds ordinary accretion, why galaxies rotate as they do without invoking dark matter, where energy goes as space expands or compresses, what a collapsing core actually becomes, and why gravity is so much weaker than every other force (H3–H9) – and, deeper still, against the question of why spacetime has any elasticity to resist curvature with at all, rather than that stiffness simply being assumed (H10). Not to replace general relativity, but to see how far "space as a real medium" can be pushed before the numbers themselves say otherwise. Several of these hypotheses do not survive their own numeric check and are withdrawn here rather than argued around (§7); what remains is offered as a disciplined set of working hypotheses, each stated plainly enough to be falsified, not as a finished alternative to general relativity.

At a glance.

Hypothesis	Question it answers	Claim, in one line	Status	Where
H1	What sets the amount of curvature for a given mass?	Gravity's strength is tied to the elasticity of space; bend amount $\propto$ elasticity, with elasticity constant	*Kept*, as a scoped recovery of GR, not independent content	§4.1
H1b	Does elasticity ever vary, and could that restore shape-sensitivity that a constant elasticity forbids?	Elasticity may depend on local field conditions (the irregularity scalar $\mathcal{I}$), not just differ between bodies – speculative, and must vanish wherever constant elasticity is already confirmed	*Open*, speculative; needs to be written as the field-equation amendment it requires	§4.2
H2a	What causes gravity, and why does it have the direction and sign it does?	Gravity is the felt consequence of spacetime curvature itself: curvature connects neighboring points to slightly different local rates of clock flow, and the widening of that gap as curvature deepens *is* the acceleration ($a = -\partial_r \Phi$, the Newtonian limit of the geodesic equation) – this paper's defining statement of gravity, consistent with the GPS correction, independent clock data, and Earth's tidal gradient, all three of which fix the same single $\Phi = -GM/r$ rather than confirming it three separate ways	*Kept*, as the defining recovery; not a result beyond GR	§4.3

Hypothesis	Question it answers	Claim, in one line	Status	Where
H2b	(alternative driver, considered and set aside)	A constant spin of space about an axis was considered as a second route to the same felt sensation; it reproduces the sensation (via the equivalence principle) but carries zero true, frame-independent curvature (a rotating frame is flat spacetime in disguise), so it does not meet this paper's own definition of gravity above – it demonstrates what gravity is *not*, not a second way gravity can arise	*Kept*, only as the flat-spacetime counterexample	§4.3
H2c	Is the spatial bend H1 computes itself a force, or only a geometric redirection?	Bend alone only redirects a moving body's direction, at constant speed; it cannot start motion from rest and is $\sim v^2/c^2$ too weak to source measured gravity at ordinary speeds – the real accelerating mechanism is the radial, g_{00}-sourced term H2a relies on (loosely called "centripetal" in this paper; see §4.3 for why that word is imprecise here)	*Kept*, standard geodesic-equation content	§4.3
H3	Why do some black holes (supermassive) far exceed the mass stellar collapse can produce?	Space is not smooth; large tears/inconsistencies in its fabric, from uneven cosmic expansion and turbulence, form the largest black holes directly	*Narrowed*: kept only as a restatement of the open seed problem, not as literal tears	§4.4
H4	What seeds the accretion that forms stars, planets, and asteroids?	Small, widespread tears/inconsistencies in space trap matter and seed accretion; every body in space has a black hole at its center	*Withdrawn*: literal claim contradicted by interior-structure data	§4.5
H5	What, specifically, are supermassive black holes, if not grown stellar remnants?	Supermassive black holes are large vortices formed where oppositely-expanding regions of space collide; they are not, in general, of stellar-collapse origin	*Narrowed*: kept only as a restatement of the open seed problem, not as literal collisions	§4.6

Hypothesis	Question it answers	Claim, in one line	Status	Where
H6	Why do stars at a galaxy's edge orbit faster than visible mass alone predicts?	A galaxy is matter settled into the depression around a central vortex (the SMBH); the vortex's strength and reach set the rotation curve directly, so dark matter is not needed	*Withdrawn*: scale mismatch (3–4 orders) is decisive as stated; target question remains open	§4.7
H7	Where does energy come from (or go) when space expands or compresses?	Expansion/compression of space is itself a loss or gain of energy, carried in part by particle creation from the vacuum – the known mechanisms involved (cosmological particle creation, the dynamical Casimir effect, Hawking radiation) are conversions, not net sources, so the claim here is unification of those mechanisms under θ, not that this energy is extractable	*Kept*, with the extraction claim explicitly dropped	§4.8
H8	What actually happens, physically, as a collapsing core becomes a black hole?	The core sinks into space gradually (not instantly), becoming more a feature of time than of space as it goes, generating a vortex; it ends as a dense, finite core, not a literal point	*Kept*, as the strongest hypothesis here; two items (angular momentum, core scale) still open	§4.9
H9	Why is gravity so much weaker than every other force?	Gravity's measured weakness (and the tiny strain of gravitational waves) is dilution: gravity's true strength lives in a hidden extra dimension, and what we measure is leakage into our own	*Kept*, only as a citation of an existing, already-bounded program	§4.10
H10	Why does spacetime have elasticity (Σ_0) at all, rather than it being a bare, assumed constant?	Spacetime's stiffness is *induced*: the aggregate residue of quantum vacuum fluctuations of every matter field, the way a liquid's surface tension is the aggregate residue of molecular cohesion, not a fundamental ingredient of its own	*Open, honestly failing*: Standard-Model fields and scale give a number $\sim 10^{30}$ off; likely the same root cause as H9's hierarchy problem	§4.11

Background and motivation. [PLACEHOLDER]

Thesis. *Treating space as an empty, static backdrop to events – rather than as a real medium with its own fluidity and elasticity – is this paper's central departure from the stan-*

dard picture, and is proposed here as a more fruitful place to locate the open problems this paper surveys than any single one of those problems taken alone. This is a reinterpretation under test, not yet an established mechanism: where the analogies used to develop it (an elastic medium, a flowing river, a honey-like fluid under load) break only against other analogies, or against a limit no one has yet checked, those breaks are read here as marking where the medium's own properties (most centrally, whether its elasticity is a true constant or responds to local conditions, H1/H1b) are not yet pinned down by observation or experiment – not as grounds to discard the picture outright. But where a specific claim built on the picture is checked against its own stated numerical prediction and fails by many orders of magnitude or is directly contradicted by existing data – as with H2a's original cosmic-expansion engine, H4's literal claim, and H6's scale claim (§7) – that is not an analogy breaking; it is a claim being measured and found wrong, and this paper withdraws that specific claim outright rather than keeping it on the books as unfinished work. The distinction that matters throughout is between the research program (space as a real, elastic medium, kept) and any one claim made under it (kept, narrowed, or withdrawn on its own numerical merits). Closing the program's remaining gaps is accordingly treated as outstanding work; closing the gaps in a specific, already-failed claim is not.

Working postulates. This paper's hypotheses share three working postulates about the nature of space, stated here explicitly so the reader can see exactly what is being assumed before any result is derived from it.

P1 (space as a medium). Space is not merely a bookkeeping device for the distances between objects; it possesses real structure of its own. This is not an unprecedented claim: quantum field theory already treats the vacuum as having real energy and structure, confirmed in the laboratory via the Casimir effect (§3.2). **Qualification, stated as plainly as the postulate itself:** P1 does *not* reintroduce a preferred, detectable rest frame for this medium. The 19th-century aether hypothesis proposed exactly such a frame, and it was ruled out by the Michelson–Morley experiment and by over a century of increasingly precise successors, now constraining any such effect to below roughly one part in 10^{17} (§4.1, §4.8 discussion). The medium P1 refers to is modeled on the quantum vacuum's energy and structure – the same in every inertial frame – not on the mechanical, frame-bearing aether that experiment has already excluded. Any development of P1 that reintroduces a detectable preferred frame takes on the burden of directly answering that body of null results; this paper does not do so, and flags this as the single most important constraint on how P1 may be developed further.

P2 (fluidity and elasticity). This medium is not rigid or inert: it is capable of expansion, compression, shear, and rotation (*fluidity*, formalized as the kinematic triplet $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$), and it resists bending by an amount that need not be assumed constant (*elasticity*, formalized as the compliance field Σ). Both are defined precisely in §3.2. **On the fluid analogy's proper scope.** "Fluid" should not be read as choosing between a continuum (liquid-like) picture and a particle (gas-like) one – quantum fluids are not required to pick one. Superfluid helium is simultaneously a continuous, frictionless medium at long wavelength *and* a gas of discrete, countable quasiparticles (phonons) at the level of its own excitations; this is exactly why superfluids and Bose-Einstein condensates are used as laboratory analogues for curved spacetime (§2, analogue gravity). If space is a real medium with quantized excitations, it should be expected to show the same duality light already shows: a gravitational wave as the continuum description, a graviton as its particle counterpart – the still-undetected, but mainstream and actively pursued, quantum of the gravitational field. Whether space genuinely has this structure remains (c) open; that a quantized medium *can* have it simultaneously is (a) established physics, not a metaphor stretched to fit.

P3 (inconsistency). This medium is not perfectly smooth: it admits departures from smoothness, formalized as the irregularity scalar $\mathcal{I}$ (§3.2) and classified by species in the taxonomy of §4.12 – holes, tears, cracks, faults, channels, and valleys, alongside the classical spherical black hole. The qualitative picture this draws on is the observed cosmic web, where matter is seen gathering along filaments and voids rather than sitting uniformly distributed, much as water gathers into rivers and lakes along the low places in a landscape. **Qualification:** this qualitative resemblance is not yet independent evidence for P3. The cosmic web's filamentary structure already has a complete, quantitative, independently confirmed explanation – gravitational instability acting on

the density fluctuations seen in the CMB, growing under cold dark matter, reproduced in detail by simulation (§4.7 discussion). For P3 to add genuine content here, rather than relabel an explained phenomenon, it needs to reproduce the same filament and void statistics from $\mathcal{I}$ and the kinematic triplet directly – the open item already flagged in the taxonomy table's *Crack/channel* row.

An illustrative picture for P1–P3. A useful way to hold P1–P3 in mind, borrowed from a standard cosmological teaching device, is the surface of an expanding soap bubble or balloon. The thin film stands in for our 3-dimensional space; its own expansion stands in for cosmic expansion; and its embedding in a further dimension, used purely to visualize that expansion without needing a center, stands in for the medium-like structure P1 proposes space itself has. Viewed from *within* the film – the only vantage any observer in it actually has – every point sees every other point receding, symmetrically, with no special point singled out as the source of the expansion; this is also exactly what real observers see from any real galaxy, in any direction, and is the central lesson the analogy is built to convey. Because light has had a finite time (on the order of 13.8 billion years) to travel since the hot Big Bang, any observer within the film sees only a limited patch of it – the true extent of the full film, and of the full universe, remains unseen and may be far larger than what is visible.

One correction the analogy is specifically built to enforce, not to blur: there is no Big Bang *point*, in the film or in the embedding dimension. The embedding direction (here identified with time) does not locate a position somewhere else in space that the expansion came *from* – it locates a *moment*: the limiting case, run backward, where the film's radius shrinks to zero and every point presently in the universe coincides at once. The Big Bang happened everywhere in space, simultaneously, at one moment in the past common to every point – not at one place, reachable in some direction, even an extra-dimensional one. Treated this way, the analogy is a correct, standard illustration of P1–P3's picture of space, rather than a restatement of the single most common misconception the analogy was devised to correct.

Every hypothesis in §4 is best read as a specific proposal for *how* P1–P3 cash out quantitatively, not as independent claims – H1 proposes a form for Σ; H2, H5, and H6 propose sources and consequences for the kinematic triplet; H3, H4, and H5 propose what $\mathcal{I}$ looks like at different scales; H7 proposes what P2's dynamics exchange with real, created energy. Judged on their own, P1–P3 are postulates, neither true nor false until their consequences are worked out and checked; judged together with the hypotheses that follow, they are the common root the taxonomy table (§4.12) and the open items throughout §4 are asking to be filled in.

A guiding conviction. Beyond P1–P3 as stated technical postulates, this paper is written from a conviction held by its author: that treating space as empty nothingness, with geometry alone doing every explanatory job, is a deeper source of several of the open problems surveyed in this paper – not only the ones P1–P3 touch most directly – and that treating space instead as a real, fluid, elastic substance is more likely to resolve those problems than to create new ones. This conviction is the motivation behind the choice of postulates and hypotheses in this paper; it is stated plainly here *as* a conviction, not smuggled in as an established result, because the two are easy to conflate and should not be. A guiding conviction earns its place in physics the same way every idea in this paper has been asked to: by being turned into specific, checkable claims and surviving contact with what is already measured – which is the task the rest of this paper, and the open items left throughout it, are still working through rather than a step this paragraph can shortcut.

Epistemic stance: inference to the best explanation, not refutation. Every hypothesis that follows is offered as an entrant in a specific kind of competition, not as an attempt to prove general relativity wrong or to establish an alternative beyond doubt. The method this paper follows is *abductive* – inference to the best explanation (Peirce; formalized by Harman, 1965, and developed at length by Lipton, *Inference to the Best Explanation*, 1991/2004) – rather than deductive refutation. The starting point, in every case, is a phenomenon already observed and already in need of *some* explanation (flat rotation curves, the supermassive black hole seed problem, gravity's hierarchy-problem weakness, and so on); the question this paper asks is not "can the standard explanation be disproven" but "among the live explanatory candidates already on offer for this specific phenomenon – including the standard explanation itself – does treating space as a real, structured medium (P1–P3) supply a candidate that fits at least as well, and in some cases arguably better, without discarding anything the standard explanation already gets right." Four things follow from stating the method this way, rather than leaving it implicit:

1. **The comparison is always among specific, named rivals, not against an undefined status quo.** Each hypothesis's foundational assessment (§2) identifies the mainstream candidate(s) it is actually competing against – direct-collapse and heavy-seed models for H3, MOND for H6, the ADD and Randall–Sundrum models for H9, and so on – so that "fits the puzzle better" is judged against a stated alternative, not against silence.
2. **Standard general relativity is not one candidate among equals; it is the reference every candidate here must already reduce to or match wherever it has been tested.** Abduction compares candidates on their explanatory merits for what is *not yet* settled; it does not reopen what already is. Where this paper's hypotheses reproduce a GR result exactly (H1's inverse-square recovery; H2c's light-bending and perihelion figures), that is table stakes for staying in the competition at all, not a point of novelty.
3. **Losing a specific comparison narrows a hypothesis; it does not indict the research program.** This is the same distinction already drawn in the Thesis above between the program (P1–P3, kept) and any one claim made under it (kept, narrowed, or withdrawn on its own merits) – stated here as the general methodological principle that distinction is an instance of. H4's literal claim and H6's scale-mismatched mechanism are withdrawn or narrowed for exactly this reason: on a fair comparison, a better-fitting candidate (ordinary accretion physics; the standard dark-matter halo) already exists, and abduction's verdict in that specific case is accordingly against this paper's entrant – a result the method is built to produce when warranted, not a failure of the method.
4. **The standard for a good fit is comparative and multi-criterial, not binary.** Where a hypothesis is kept or narrowed rather than withdrawn, the judgment rests on the same few questions asked of any explanatory candidate: does it remain consistent with everything already measured; does it reduce, rather than multiply, the number of separately-adjustable assumptions needed; does it reach at least as far observationally as its named rival; and can it be stated precisely enough to fail a future test. None of these, singly or together, constitute proof – abduction never yields that – and this paper does not claim otherwise anywhere in what follows.

Read this way, the tier labels used throughout §2 and §4 – *standard*, *extrapolated*, *speculative*; *kept*, *narrowed*, *withdrawn* – are not verdicts of proof or disproof. They are this paper's own abductive bookkeeping: a record of which puzzle-piece candidate currently fits best, for which puzzle, against which named rival, revisable the moment a better-fitting candidate – standard or otherwise – is identified.

A thought experiment: the ant on water. A single worked example fixes, with an exact number rather than only an analogy, what this paper means whenever it says a governing equation can hold unchanged across scales while still looking, practically, as if different rules applied at different sizes.

The experiment. An ant walks, unharmed, across a puddle of spilled water. The same water, as a lake, drowns anything that falls in; as open sea, it does so even more readily, and raises much larger waves while it is at it. The water's molecular identity – the same H_2O, the same intermolecular forces – never changes across these three cases. What changes is which of two forces, both always present and both governed by fixed, scale-independent laws, happens to dominate at the size in question.

The resolution. Surface tension acts along a contact perimeter and so scales with *length*, $F_{\text{tension}} \sim \gamma L$; weight scales with *volume*, $F_{\text{weight}} \sim \rho g L^3$. Because weight falls off as the cube of size and tension only as the first power, there is a single, exactly calculable crossover length – the capillary length –

$$\ell_c = \sqrt{\frac{\gamma}{\rho g}} = \sqrt{\frac{0.072 \text{ N/m}}{(1000 \text{ kg/m}^3)(9.8 \text{ m/s}^2)}} \approx 2.7 \text{ mm},$$

below which surface tension always wins and above which gravity always wins, for *exactly* the same water, governed by exactly the same γ, at every scale. An ant's leg contact points sit at this scale (as do water-striders', independently confirmed); nothing the size of a lake or a human ever can, however calm the water, because ℓ_c itself does not move. The separate fact that waves grow from puddle to lake to open sea is a different effect again – set by wind fetch, not by any change in the water's own bulk law – and is kept distinct here deliberately, so the one lesson this example is built to carry is not blurred by a second, unrelated one.

The general lesson, stated precisely. A scale-dependent crossover in which term of a fixed governing equation dominates is evidence that the equation holds unchanged across scales, not evidence against

it. "The rules seem to change with scale" and "new terms, present but negligible elsewhere, come to dominate at this particular scale" are observationally the same statement; this paper deliberately adopts the second phrasing throughout, because it is the one that keeps the underlying law intact rather than discarding it. The same structure recurs, at much greater stakes, twice already in this paper: Newtonian gravity is not falsified by general relativity, it is the $v \ll c$, $GM/rc^2 \ll 1$ limit of the same geodesic equation (§4.3); and ordinary quantum superposition is not falsified by the classical behavior of everyday objects, it is suppressed by *environment-induced decoherence* (Zel'dovich & Zurek; see Zurek, *Rev. Mod. Phys.* **75**, 715 (2003), and Joos & Zeh, 1985) – entanglement with a large number of environmental degrees of freedom that grows overwhelmingly effective as an object's size and interaction count increase, not a breakdown of the Schrödinger equation itself. Three instances of one pattern: the underlying law is untouched in every case; what changes is which term, among several always present, happens to be the one an observer at a given scale actually notices.

What this licenses for this paper, and what it does not. This is the template every elasticity- and defect-scale claim in §4 is implicitly asking to satisfy: a term, present in the governing equation everywhere, that is negligible across the entire range already tested by GR's confirmed results (§2) and becomes significant only outside that range – near the Planck scale for H1/H10, near cosmological horizon scales for H3/H5 (§4.1, §4.4, §4.11). The ant's crossover is licensed by three independently and precisely measured constants (γ, ρ, g); this paper's analogous crossover scales are not yet pinned down to anything like that precision, and this thought experiment is offered here only as the standard the rest of the paper is trying to eventually meet, not as evidence already in hand that spacetime possesses such a term. Where a specific hypothesis's foundational assessment (§2) says its crossover scale remains unspecified, that is the gap this example is pointing at.

Scope and disclaimers. [PLACEHOLDER]

2 Foundational Basis and Prior Work

2.1 Established results this paper relies on

- **Equivalence principle** (Einstein, 1907/1911; tested to high precision by Eötvös-type and modern torsion-balance experiments): locally and instantaneously, a uniformly accelerated frame is indistinguishable from a uniform gravitational field. H2's original engine used this unchanged, but extended its scope from a local, instantaneous statement to a claim about the global *origin* of gravity – an extension that the magnitude checks in §4.3 rule out, and that engine is retired for exactly that reason. H2-revised, kept, does not make this extension at all: it uses the equivalence principle only in its ordinary, local scope, as the standard justification for treating g_{00}'s gradient as the gravitational acceleration.
- **Einstein field equations**, $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}/c^4$: curvature is set by mass-energy through the fixed constant G. H1 keeps this structure but reinterprets G (or the coupling it plays) as a measurable elasticity rather than a bare constant – a reinterpretation of an existing equation, not a new one, until "elasticity" is given independent empirical content.
- **Measured constancy of G**: Newton's constant is measured to be constant across laboratory tests to good precision. Any version of H1 in which elasticity varies must explain why that variation has not already shown up in existing G measurements.
- **Cosmic expansion and its acceleration** (Hubble's law; the accelerating expansion inferred from Type Ia supernovae, Riess et al. 1998, Perlmutter et al. 1999): real, but measured to be extremely small and essentially uniform on the relevant scales – not currently known to be locally concentrated around individual masses the way H2a would need.
- **Observed isotropy of the cosmic microwave background**: once the known dipole (from our own solar system's motion, ∼368 km/s) is subtracted, the residual anisotropy is at the $\Delta T/T \sim 10^{-5}$ level. This bounds a specific kind of claim very tightly – a single, coherent, *global* rotation or directional stretching of the universe as a whole, fit via Bianchi-type anisotropic cosmological models against this residual (e.g. Jaffe et al., using WMAP), giving roughly $\omega/H_0 \lesssim 10^{-9}$. It does *not* bound local, patchy inhomogeneity – the 10^{-5} residual itself *is* exactly that local texture, and

is the measured seed of all structure. Directly bearing on H2b, whose claim (a single, universe-wide spin) is precisely the global kind this bound constrains; does not by itself bear on P3's local-irregularity premise or on a patchy (rather than globally directional) reading of H5.

- **Smoothness (differentiability) of the spacetime manifold**: a foundational assumption of classical GR; a literal topology change (a "tear") is excluded by it. Bears directly on H3/H4.
- **Mass bound on stellar-collapse black holes** (progenitor mass limits; the pair-instability mass gap, roughly 50–150 $M_\odot$): well established, and correctly used as the starting observation for H3.
- **Observed supermassive black holes at high redshift**: the "seed problem" – a genuine, open question H3 correctly identifies, already addressed in the literature by direct-collapse, heavy-seed, and super-Eddington-accretion proposals.
- **Helioseismology, terrestrial seismology, and planetary geodesy**: direct observational probes of the interiors of the Sun, Earth, and other bodies, bearing directly on H4's claim that a black hole sits at the center of every body.
- **Quantum foam** (Wheeler, 1955) and **primordial black holes** (Zel'dovich & Novikov, 1966; Hawking, 1971): existing, legitimate research programs nearest in spirit to H3/H4's "inconsistencies in space," discussed below.
- **Homogeneity and isotropy of cosmic expansion (the FLRW model)**: supported by CMB isotropy and large-scale-structure surveys; the expansion rate is a single scale factor, not a directional flow field with distinct colliding regions. This is a statement about the universe's *global* expansion having no preferred direction or coherent flow pattern – it is fully compatible with, and indeed requires, the same local 10^{-5}-level density patchiness noted above. Bears directly on H5's specific claim of large, coherently *directional* colliding regions; a reading of H5/H3 built instead on scattered, local patchiness (consistent with P3) is not what this bound constrains, and would need to be checked instead against GR's smoothness assumption (next item) and against whether the known 10^{-5} perturbations, via ordinary gravitational instability, already account for the structure in question.
- **Black hole rotation, frame dragging, and analogue-gravity vortex experiments** (Kerr metric; Lense–Thirring effect, now measured via accretion-disk spin diagnostics; Unruh, 1981, and later draining-fluid-vortex demonstrations of superradiance): the genuine, if indirect, point of contact for H5's vortex picture.
- **Galaxy rotation curves** (Rubin & Ford, 1970s onward): the original, and still central, dark-matter evidence H6 targets.
- **Independent dark-matter evidence beyond rotation curves**: the Bullet Cluster's lensing/gas offset; the relative heights of the CMB acoustic peaks; the growth rate of large-scale structure. Each is logically independent of rotation curves and must be addressed by any replacement, not just the curves themselves.
- **Modified Newtonian Dynamics (MOND)** (Milgrom, 1983): the existing, well-studied alternative to dark matter closest in spirit to H6, with known successes (individual rotation curves) and known difficulties (cluster scales, the Bullet Cluster, the CMB).
- **Cosmological particle creation** (Parker, 1968), **inflationary reheating**, the **dynamical Casimir effect** (Wilson et al., 2011), and **Hawking radiation** (Hawking, 1974): four established mechanisms by which a changing spacetime or vacuum boundary converts vacuum fluctuations into real particles – the direct precedent for, and current scope of, H7.
- **The Chandrasekhar and Tolman–Oppenheimer–Volkoff limits**; **the river model of gravity** (Painlevé, 1921; Gullstrand, 1922); **the interior Schwarzschild solution's space/time coordinate swap**; and **Planck stars** (Rovelli & Vidotto, 2014) and **fuzzballs** (Mathur, 2005): five established or seriously-studied results underlying H8, covering, respectively, why collapse becomes unstoppable, how space "flows into" a mass, why infall resembles motion through time, and what might replace a literal singularity.
- **The hierarchy problem** (gravity is measured to be roughly 10^{36} times weaker than electromagnetism between two protons, with no settled explanation for the gap), **large extra-dimension (braneworld) models** (Arkani-Hamed, Dimopoulos & Dvali, 1998; Randall & Sundrum, 1999),

and the **sub- millimeter torsion-balance tests of Newton's inverse-square law** (the Eöt-Wash experiments) that already bound any such extra dimension to below roughly 40–50 μm: the real, quantified starting point and the real, quantified constraint for H9.

2.2 Where each hypothesis is standard, extrapolated, or speculative

H1 (elasticity of space), scoped version. The claim that mass-energy curves space, and that the amount of curvature follows from a fixed coupling, is (a) standard GR. The claim that this coupling should be understood as a physical *elasticity* is (b) an extrapolation: it has real precedent as an analogy (spacetime-as- elastic-medium pictures appear in induced-gravity and analogue-gravity literature, §2.3 below) but is not itself an established, independently measured quantity. As now scoped (§4.1) – Σ held constant, bend computed from mass and distance via the shell theorem, and the question of *why* bending happens explicitly deferred to H2 – H1's defensible core is (a) standard: it is a direct, checked recovery of Newtonian surface gravity (confirmed numerically for Earth, Mars, the Sun, Jupiter, and Saturn) in elasticity vocabulary, with no claimed deviation from $G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$. It remains (c), speculative and currently unfalsifiable on its own, only in the direction it has not yet been developed: a specific, independently measurable variation of Σ (provisionally H1b, tied to the irregularity scalar $\mathcal{I}$) would be needed before H1 makes a prediction standard GR does not already make.

H1b (variable elasticity tied to local field conditions). That a locally-responsive coupling could restore the shape-sensitivity a constant Σ forbids is (b) a reasonable extrapolation of H1 – it is the one lever within this paper's own formalism that could do so without contradicting anything already confirmed about Σ not differing between bodies. As stated, with no function h and no coupling λ specified, it is (c) fully speculative and not yet falsifiable. It inherits a real, numerical constraint directly from its nearest precedent regardless: scalar-tensor gravity's existing solar-system bound ($\omega > 40{,}000$) already tells us how small any such effect must be wherever $\mathcal{I}$ is small, which is a concrete, borrowed falsifiability anchor even before H1b specifies its own function.

H2 (gravity sourced by expansion or spin) – original framing, since revised. This assessment applies to H2's *original* statement, kept here for the record of why it was revised rather than as the hypothesis's current form (see §4.3 for the revision this paper now adopts as its defining statement of gravity). The equivalence principle itself is (a) standard and extremely well tested. Using it to say that *a* sustained acceleration feels like gravity is correct. Using it to say that cosmic expansion or cosmic rotation *is the acceleration that constitutes all gravity, everywhere, including the mass-centered field around the Earth* is (c) a new postulate with no direct precedent and, as detailed in §4.3, was in tension with the measured size and uniformity of cosmic expansion, with CMB-isotropy bounds on cosmic rotation, and with the basic mass-centered, inverse-square shape of every gravitational field actually observed. This was the weakest part of either hypothesis in its original form, precisely because it was the most novel claim and the one most directly checked against existing, precise observations – which is exactly why H2a was revised to locate the cause in curvature itself rather than in an external driver, removing the objection rather than carrying it forward.

H3 (tears as the origin of supermassive black holes). That stellar-collapse black holes are mass-limited, and that supermassive black holes pose an unsolved seed problem, is (a) standard and accurate. That the resolution is a literal tear in spacetime's fabric is (c) a new postulate: GR's manifold is smooth by construction, so this is not an extrapolation of an existing pattern but a proposed exception to a foundational assumption, and it currently has to compete with, rather than build on, the mainstream candidate answers (direct collapse, heavy seeds, super-Eddington growth) without yet stating what distinguishes it from them observationally.

H4 (small inconsistencies seeding all accretion; a black hole at the center of every body). That accretion builds stars, planets, and asteroids, and that growing mass deepens a gravitational well, is (a) standard accretion physics, just stated in this paper's own vocabulary. That the *trigger* for accretion is a small tear or hole in space, rather than ordinary gravitational instability in a gas or dust cloud, is (c) a new, currently unnecessary postulate – the mainstream mechanism already accounts for the observation without it. The claim that every body has a black hole at its center is also (c), and is the most strongly constrained part of either hypothesis: it is checked directly against existing interior-structure data (helioseismology, seismology, geodesy) and, as stated without qualification, does not currently match it.

H5 (supermassive black holes as colliding-space vortices). The starting observation – that

a single star's collapse cannot directly produce a supermassive black hole – is (a) standard and correctly stated. The vortex/rotation imagery has (b) a real, extrapolatable point of contact: real black holes do drag spacetime when they rotate, and that rotational mathematics is exactly what analogue-gravity vortex experiments exploit. But the specific sourcing mechanism – large regions of space expanding in opposite directions and colliding – is (c) a new postulate with no observational counterpart in the standard, homogeneous-and-isotropic picture of cosmic expansion, and it has to explain away, not just outpace, the mainstream seed-growth picture (stellar-mass seeds plus mergers and sustained accretion), which already accounts for supermassive black holes' observed mass and spin without requiring colliding expansion.

H6 (galaxies as vortex depressions, replacing dark matter). The target observation – flat rotation curves – is (a) standard and correctly identified. Attributing it to the central vortex (the galaxy's supermassive black hole) is (c) a new postulate that, as it stands, faces a direct scale problem: the black hole's gravitational reach is three to four orders of magnitude smaller than the radii where flat rotation curves are actually measured, so the mechanism as stated does not yet reach the regime it is meant to explain. It is also (c) silent on the independent, non-rotation-curve evidence for dark matter (Bullet Cluster, CMB acoustic peaks, structure growth), each of which would need its own account. This is the most empirically consequential hypothesis in the set so far, precisely because dark matter is supported by several independent lines of evidence rather than one – a replacement has to clear all of them, not just the one it was built to match.

H7 (energy from expansion/compression of space). This is the best-grounded of the seven: the general phenomenon – a changing spacetime or vacuum boundary converting vacuum fluctuations into real particles – is (a) standard, established physics in four separate, serious contexts (cosmological particle creation, inflationary reheating, the dynamical Casimir effect, Hawking radiation). Treating these four as instances of one general principle tied to θ is (b) a reasonable extrapolation – a unification claim, not a new physical mechanism. Where H7 becomes (c), a new and much harder claim, is in the inference that this energy "should be harnessed": every established instance is a conversion process with a specific, identifiable cost paid elsewhere (the black hole's own mass, the energy used to drive a boundary at relativistic speed, the inflaton field's energy), not a net source, so the extraction claim is not an extrapolation of the established pattern but a departure from it, and currently has no proposed mechanism stating what would pay its cost.

H8 (black hole formation as a core sinking into time). This is the best-grounded hypothesis in the set. "Too heavy to be supported" is (a) standard stellar-collapse physics outright. "Space flows into the mass, carrying matter with it" is (a) a correct restatement of the already-standard river model. "Infall resembles motion through time" is (a) a precise, correct description of a known feature of the Schwarzschild interior, not a metaphor stretched past its proper use. "A finite core, not a literal point" is (b), at best a specific proposal competing with – or possibly a restatement of – existing, actively studied quantum-gravity candidates (Planck stars, fuzzballs) for the same open question, rather than a wholly new claim. The one piece that is (c), new and currently unaddressed, is the vortex: nothing about sinking by itself, with no stated angular momentum, produces rotation in the river-model picture, so H8 needs to add that ingredient explicitly rather than treat the vortex as automatic.

H9 (gravitational leakage into a hidden dimension). The starting observation – gravity is roughly 10^{36} times weaker than electromagnetism, with no settled explanation – is (a) a real, established, unsolved puzzle (the hierarchy problem). That this weakness could be dilution into an extra spatial dimension is (b) a direct extrapolation of existing, actively studied large-extra- dimension models (Arkani-Hamed, Dimopoulos & Dvali, 1998; Randall & Sundrum, 1999), not a new mechanism invented for this paper. As stated, with no value given for n (the number of extra dimensions) or ℓ_* (the fundamental length scale), H9 is (c) not yet falsifiable on its own terms, and it is already significantly constrained by existing data regardless of how it is specified: sub- millimeter torsion-balance tests already bound any such dimension to below $\sim$40–50 μm, which rules out the more dramatic readings of "huge in 4D" and leaves only a small, tightly bounded version available.

2.3 Relation to existing programs

H1 sits closest to *induced gravity* (Sakharov, 1967), where G itself is proposed to emerge from vacuum fluctuations rather than being fundamental, and to various *analogue-gravity* treatments that model spacetime's response to mass with elastic-medium mathematics as a teaching or effective-theory device. These analogue treatments are not chosen arbitrarily: *superfluid vacuum theory* and related programs use quantum fluids (superfluid helium, Bose-Einstein condensates) specifically because such fluids are

simultaneously a continuum medium *and* a gas of quantized excitations (phonons) – the same duality light shows as a wave/photon pair, and the same duality a quantized gravitational field would show as a gravitational-wave/graviton pair, were gravity's quantization ever confirmed. This sharpens, but does not resolve, H1's elasticity claim: it says a real quantum medium *can* consistently be both fluid-like and particle-like at once, not that space *is* such a medium. Verlinde's entropic-gravity proposal (2011) is a different but related example of treating gravity as emergent from a coarser underlying layer rather than fundamental. None of these establish that spacetime literally possesses a material elastic modulus; they are precedents for the *style* of H1's claim, not confirmations of it. **H1b** specifically sits closest to *scalar-tensor gravity* (Brans–Dicke, 1961), which already promotes G to a dynamical field and is tightly bounded in the solar system ($\omega > 40{,}000$ via Cassini Shapiro-delay tracking, Bertotti, Iess & Tortora 2003) – the nearest existing, numerically precise precedent for how small any field-dependent variation in Σ must already be wherever it has been tested; and, for the style of a locally-triggered departure from constant coupling, to *MOND* (Milgrom, 1983, §2.3 below), though MOND's trigger is a universal acceleration scale, not the irregularity scalar $\mathcal{I}$.

H2's "sinking into space" picture has a rigorous existing counterpart worth knowing about: the *river model of gravity* (Painlevé, 1921; Gullstrand, 1922; popularized by Hamilton & Lisle, 2008), in which space itself is pictured as flowing inward toward a mass – mathematically just a coordinate choice (Painlevé–Gullstrand coordinates) fully equivalent to standard Schwarzschild/GR, not a modification of it. That model gives "mass makes space flow/sink" a precise, already-solved form, but notably sources the flow from the *local* mass, not from cosmic expansion or universal spin – the opposite of what H2a/H2b propose. A genuine, non-trivial rotating universe is also an existing GR solution (Gödel, 1949), but it is non-expanding, permits closed timelike curves, and is not considered a model of our universe; current observations bound our universe's actual rotation far below what H2b would require. Distinguishing H2 from the river model – i.e. stating clearly whether the "sinking" is sourced locally (as in the river model) or globally (as currently written) – is the single most useful next step for sharpening H2.

H3 and H4 sit closest to two existing, separate programs that this paper's single notion of "inconsistency" currently blends together. *Quantum foam* (Wheeler, 1955) proposes that spacetime is topologically non-trivial at the Planck scale due to quantum fluctuations – a real precedent for "space is not smooth," but at a scale enormously below anything relevant to black-hole formation, with no established bridge to that scale. *Primordial black holes* (Zel'dovich & Novikov, 1966; Hawking, 1971) propose that density fluctuations in the early universe collapse directly into black holes of a wide range of masses, independent of stellar collapse, and have been studied both as dark-matter candidates and as possible seeds for structure formation – a real precedent for "small holes seed accretion" and even for "some black holes are not stellar remnants," but sourced by early-universe density fluctuations, not by tears from cosmic expansion or turbulence. Reframing H3/H4 explicitly in terms of one of these two existing programs – rather than the new term "inconsistency," which currently has no agreed definition in either GR or quantum gravity – would let the hypotheses inherit an existing, worked-out mathematical machinery instead of having to build one from scratch.

H5 sits closest to two existing, separate programs as well. *Analogue gravity* (Unruh, 1981, and subsequent draining-vortex experiments) shows that a real fluid vortex can mathematically mimic a rotating black hole's horizon and ergosphere for waves travelling in that fluid – a precedent for the mathematics of a "vortex," but one built because sound waves in a draining fluid obey the same wave equation as fields near a Kerr horizon, not because spacetime itself is a fluid that forms vortices. *Bubble-collision black-hole formation* during an early-universe phase transition (studied in certain inflationary and phase-transition cosmological models) is the closest precedent for "a collision forms a black hole," but it is a mechanism specific to the very early universe and to a particular particle-physics scenario, not a general or ongoing process tied to cosmic expansion as such.

A third, more specific precedent is worth adding alongside these two: *brane cosmology*, and in particular the *ekpyrotic/cyclic universe* models (Khoury, Ovrut, Seiberg, Steinhardt, 2001; Steinhardt & Turok, 2002), in which our observable universe is a 3-dimensional "brane" embedded in a higher-dimensional bulk, and the hot Big Bang itself is proposed to be the literal collision of our brane with another one moving through that extra dimension. This is the closest existing precedent to "colliding space" taken literally, as a worked, higher-dimensional mechanism rather than a metaphor – note that it requires a dimension *beyond* ordinary 4-dimensional spacetime (3 space, 1 time), not simply a fourth dimension, since spacetime already has one. Two caveats temper it as support for H5 specifically: (i) it is a mechanism proposed for a single, universe-defining event – the origin of the Big Bang itself – not a repeating, local process that could separately spin up a vortex at the center of each galaxy over the course of cosmic history, so H5 would need its own argument for extending a one-time origin mechanism

into a recurring, local one; and (ii) ekpyrotic/cyclic models remain a minority position within cosmology, disfavored by some researchers relative to standard inflation, in part over specific predictions (such as the pattern of primordial gravitational waves) that current data has not decisively settled either way.

As with H3/H4, stating which of these three programs H5 means to extend – and on what grounds its collision mechanism applies well beyond the regime (early-universe-origin, or a fluid-analogy for waves) where its nearest precedents are actually set – is the clearest way to sharpen it.

H6 sits closest to *MOND* and its relativistic extensions (TeVeS and successors), which already pursue exactly H6's stated goal – explaining rotation curves without a new matter species – by modifying the force law or the field equations at low accelerations, rather than by invoking a vortex. MOND's real track record is informative either way: it fits individual galaxy rotation curves remarkably well, which is evidence that *some* modification-of-gravity approach can work at that scale: but it has never comfortably accounted for the Bullet Cluster or the CMB acoustic-peak ratios without reintroducing some form of unseen mass. If H6 is pursued further, inheriting MOND's success on rotation curves is the easy part; it will need its own answer to the harder part MOND has not solved, rather than treating rotation curves as the whole problem.

H7 does not need a new precedent – it names four that already exist and already agree with one another: cosmological particle creation (Parker, 1968) and inflationary reheating describe expansion converting vacuum/field energy into real particles; the dynamical Casimir effect demonstrates the same conversion from a moving boundary in the laboratory; Hawking radiation demonstrates it from a compressing (collapsing) geometry. H7's actual contribution, if developed, would be showing these four are special cases of one relation involving θ – a worthwhile, well-posed piece of theoretical unification – provided it stops there, at the unification claim, rather than extending to net extraction, which none of its four precedents support.

H9 sits directly on top of two existing, closely related programs: the **ADD model** (Arkani-Hamed, Dimopoulos & Dvali, 1998), in which gravity's weakness is explained by its spreading into n large extra dimensions while the other forces stay confined to our 4D brane, and the **Randall–Sundrum model** (1999), which achieves a similar dilution with a warped, rather than flat, extra dimension. Both were developed specifically to address the hierarchy problem H9 starts from, so H9 is not proposing a new mechanism, only adopting an existing one and connecting it to this paper's own language (Σ_0 as the diluted, 4D-effective elasticity). Both parent models are already tightly constrained by sub-millimeter gravity tests and collider searches for missing-energy gravitons, and H9 inherits those constraints directly rather than starting from a clean slate.

3 Notation and Primitive Objects

- Manifold / spacetime: [PLACEHOLDER, e.g. a 4-dimensional Lorentzian manifold $(M, g_{\mu\nu})$ with signature $(-,+,+,+)$]
- Units and conventions: [PLACEHOLDER, e.g. $c = 1$ unless stated, G retained explicitly]

3.1 Postulate P1: Spacetime as a Physically Real Dynamical Field

Every hypothesis in §4 assigns spacetime properties – elasticity, fluidity, rotation, inconsistency – that only make sense to ask about if spacetime is the kind of thing that *has* properties, rather than a bookkeeping device for where matter happens to be. This subsection states that ontological claim once, explicitly, as P1, rather than leaving it implicit in every hypothesis that depends on it.

Statement. *Spacetime is a physically real, dynamical entity – a field, in the same ontological sense as the electromagnetic field – and not merely a descriptive geometric scaffold for locating events.*

Argument. The claim is established by a simple exclusion, not by analogy:

1. **Premise.** Whatever is nothing – a pure absence, with no evolution equation, no independent degrees of freedom, no capacity to carry energy – cannot bend, oscillate, or propagate a disturbance independently of a source. These are all actions, and nothing cannot act.

2. **Observation.** Spacetime curvature does all three: it obeys its own evolution equation (the Einstein field equations, $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}/c^4$, a genuine dynamical law, not a static definition); it propagates as a disturbance that persists and travels long after, and independently of, the event that generated it (a gravitational wave); and it transports energy while doing so – measured indirectly, for decades, in the Hulse–Taylor binary pulsar's orbital decay (Hulse & Taylor, 1975; Taylor & Weisberg, 1982), and measured directly in LIGO/Virgo's detections of merging compact objects beginning in 2015 (Abbott et al., 2016).
3. **Conclusion.** By the premise, spacetime is therefore not nothing: it is a physically real, dynamical entity.

This is a valid exclusion (modus tollens on "nothing ⇒ cannot bend, propagate, or carry energy"), and its second premise is observational fact, not interpretation. The conclusion it licenses is exactly, and only, "spacetime is real and dynamical" – the same ontological status already attributed to the electromagnetic field. It does *not* license the further, stronger claim that spacetime is a classical material medium with a rest frame or discrete phases of matter (solid, liquid, gas); that would be a second, independent claim requiring its own argument, and the next paragraph explains why this paper does not make it.

Scope: field, not aether. A classical medium carries a rest frame – the frame in which the medium itself is stationary – and a moving observer should in principle detect motion relative to it. Michelson–Morley (1887) and its modern descendants (resonant- cavity and dual atomic-clock tests of Lorentz invariance) rule out any such frame to roughly 1 part in 10^{17}–10^{18}. P1 is constructed to avoid this constraint entirely rather than fight it: it attributes to spacetime only what the argument above actually establishes (a dynamical field with its own equation of motion, capable of propagation and energy transport), which is precisely the ontological status already held, uncontroversially, by the electromagnetic field – real and dynamical, but with no preferred rest frame and no particulate or phase structure. Einstein's own 1920 Leiden address, *Ether and the Theory of Relativity*, makes exactly this distinction: general relativity, he argued, requires something "ether-like" in the sense of a physically real, spatially-extended field, but explicitly not in the older mechanical sense of a substance with a state of motion. P1 adopts that same, restricted sense, and no more.

A fourth line of evidence: frame-dragging. The three-part observational case above – evolution equation, propagation, energy transport – has a fourth member worth stating explicitly here rather than leaving it filed only under H5's supporting citations (§2): a rotating mass measurably drags the local structure of spacetime around with it, not merely curving it statically. This is the Lense–Thirring effect (1918), confirmed directly for Earth by Gravity Probe B – reporting a frame-dragging precession of 37.2 ± 7.2 milliarcseconds/year against a predicted 39.2 mas/yr (Everitt et al., 2011) – and independently, to a few percent, via LAGEOS/LAGEOS II satellite laser ranging. A gyroscope in orbit, referenced against a distant guide star, precesses simply because a rotating mass is nearby: spacetime's local structure is being pulled along, something a static bookkeeping geometry could not do, and a genuinely independent confirmation of P1 alongside the other three. **The same caveat from the paragraph above applies here with particular force, precisely because the language of "dragging" invites the reading P1 is built to exclude.** Frame- dragging is a local, relative effect – what is measured is one gyroscope's precession relative to a distant reference, fully consistent with general covariance – not evidence of an absolute medium at rest somewhere that the Earth is dragging *through*. It adds a fourth data point to "spacetime is real and dynamical"; it adds nothing to, and must not be read as reopening, the question Michelson–Morley and its successors already closed.

What P1 does and does not do for the hypotheses below. P1 is the license to ask "what are spacetime's dynamical properties?" at all – it is the reason $\Sigma(x)$ (elasticity) and $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$ (fluidity), defined next, are properties of something real rather than figures of speech applied to empty bookkeeping. P1 by itself does not supply the *values* or *functional form* of any of those properties, and it does not by itself explain *why* the Einstein field equations take the specific form they do – a field being real and dynamical is compatible with many possible evolution equations, and P1's argument used the field equations as observed fact, not derived them. That derivation – showing $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}/c^4$ as a consequence of some independently specified property of the field (elastic, thermodynamic, or otherwise), rather than as this paper's starting assumption – is left as the open problem it is elsewhere in existing research (Sakharov's induced gravity, 1967; Jacobson's thermodynamic derivation from the Clausius relation, 1995; Verlinde's entropic gravity, 2010) and, candidly, in this paper's own H1 and H10 (§4.1,

§4.11): until one of those is carried through, $\Sigma(x)$ below is a reinterpretation of an existing constant, not yet an independent derivation of it.

3.2 Core Terms: Elasticity and Fluidity of Space

Hypotheses H1–H6 all lean on two intuitive properties of space – its *elasticity* (how much it resists bending) and its *fluidity* (how it flows, expands, spins, or shears locally). Rather than inventing new mathematics for these, each is defined here as a generalization of an object GR already has, so that "standard GR" is the special case where the generalization is held fixed, and each hypothesis's new content is exactly the claim that it isn't.

Elasticity: the compliance field Σ. Standard GR writes $G_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$, a fixed coupling. The combination $c^4/(8\pi G) \approx 4.98 \times 10^{42}$ N is already informally called the *stiffness of spacetime* (Thorne and others) – it is how much stress-energy it takes to produce a given amount of curvature, i.e. exactly the quantity H1 calls elasticity, already present in the theory as a constant. Define

$$\Sigma(x) \;:=\; \frac{c^4}{8\pi G(x)}, \qquad G_{\mu\nu} \;=\; \frac{1}{\Sigma(x)}\, T_{\mu\nu}. \tag{1}$$

Standard GR is the case $\Sigma(x) = \Sigma_0$, constant everywhere. **H1's content is exactly, and only, the claim that Σ varies** – with density, curvature, scale, or something else to be specified – subject to reducing to the measured constant Σ_0 everywhere G has actually been tested (§4.1). Writing $\Sigma(x)$ rather than "elasticity" turns H1 into a statement about one specific field, with one specific job: say what Σ depends on, and by how much, and the hypothesis becomes checkable.

Fluidity: the kinematic triplet $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$. GR already has a standard, rigorous way to describe a "flow" of space: take a congruence of worldlines (e.g. the worldlines of observers at rest with respect to local matter) and decompose how nearby worldlines move relative to one another into three pieces –

- θ, the **expansion scalar**: the local rate at which a small volume of worldlines grows or shrinks. This is precisely cosmic expansion, generalized to be local: in the standard cosmological model $\theta = 3H$, three times the Hubble parameter. **H2a's "expanding space"** is a statement about θ.
- $\omega_{\mu\nu}$, the **vorticity (rotation) tensor**: the local rotation of nearby worldlines about each other. **H2b's "spin of space"** and the rotational character of **H5's and H6's "vortex"** are statements about $\omega_{\mu\nu}$ (or its vector form ω^μ).
- $\sigma_{\mu\nu}$, the **shear tensor**: local stretching or squeezing without a change in volume or net rotation – distortion into an ellipsoid rather than a sphere. This is the natural home for **H3/H4/H6's "turbulence," "tears," "cracks," and "channels"**: a shear-dominated region stretches matter preferentially along one direction, which is exactly the qualitative picture of a filament or a crack, as opposed to θ (uniform swelling) or $\omega_{\mu\nu}$ (rotation).

These three are not independent of curvature or of matter: they are tied together by the **Raychaudhuri equation**,

$$\frac{d\theta}{d\lambda} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu, \tag{2}$$

where the last term is fixed, through the Einstein field equations, by the local mass-energy density. This is the precise, already- established sense in which "fluidity" cannot float free of a mass-energy source (the open requirement raised against H2, H4, H5, and H6 throughout §2 and §4): θ, $\sigma_{\mu\nu}$, and $\omega_{\mu\nu}$ are properties of how matter moves and curves spacetime, not properties spacetime can be assigned independently of matter. **Using this formalism does not rescue any hypothesis from that requirement** – it hands you the exact equation that requirement is written in, so that "what sources this" becomes a specific term to compute rather than an open question to wave at. Stated as plainly as possible: any hypothesis that wants θ or $\omega_{\mu\nu}$ to *cause* curvature, rather than merely describe how a congruence of observers moves through a spacetime whose curvature is already fixed by $R_{\mu\nu}u^\mu u^\nu$, is proposing to change that last term – which is an amendment to the Einstein field equations themselves, not a free parameter available within them. H1b, H2, and H5 should each be read as needing to say, explicitly, which term they are amending, rather than leaving the kinematic triplet to imply a sourcing role the Raychaudhuri equation does not, on its own, give it.

Inconsistency: the irregularity scalar $\mathcal{I}$. H3–H6 all invoke a third core term – that space is not smooth, that it is "riddled with inconsistencies." This, too, already has a rigorous counterpart to generalize rather than a word to leave informal: standard cosmological *perturbation theory* already describes the real universe as a smooth background (homogeneous, isotropic FLRW) plus deviations from it, and measures those deviations with a dimensionless quantity (conventionally δ, the density contrast) relative to that background. Define, in the same spirit, an **irregularity scalar**

$$\mathcal{I}(x) \;:=\; \ell^2 \sqrt{(\theta(x)-\bar{\theta})^2 \;+\; 2\,\sigma_{\mu\nu}(x)\sigma^{\mu\nu}(x) \;+\; 2\,\omega_{\mu\nu}(x)\omega^{\mu\nu}(x)}\;, \tag{3}$$

where $\bar{\theta}$ is the smooth background expansion rate at that cosmic time and ℓ is a reference *time* scale fixed by convention – not a length, despite the suggestive symbol: θ, $\sigma_{\mu\nu}$, and $\omega_{\mu\nu}$ each carry units of inverse time, so it is ℓ^2 with units of s^2, not m^2, that is needed to make $\mathcal{I}$ dimensionless (§5's units check). $\mathcal{I}(x) = 0$ recovers perfectly smooth FLRW space; $\mathcal{I}(x) > 0$ is a "spacetime inconsistency" in H3–H6's sense, with its *magnitude* setting how large a departure it is and its *dominant term* (θ, $\sigma_{\mu\nu}$, or $\omega_{\mu\nu}$) setting which taxonomy species it looks like (§4.12). This is deliberately *not* how the Sphere species (an actual black hole) should be defined, however: GR already has an exact, independent, and far better-established criterion for that – a **trapped surface** (Penrose, 1965), a closed surface from which even outward-directed light cannot escape – and the formation of a horizon should be checked against that criterion, not against a large value of $\mathcal{I}$. Treating $\mathcal{I}$ as graded, continuous "inconsistency" for the Vortex/Depression/Crack species, while keeping the Sphere species tied to the trapped-surface criterion it already has, is what keeps the taxonomy from quietly promoting an ordinary density fluctuation into a black hole merely because some scalar got large.

What remains open: the threshold value of $\mathcal{I}$ at which a region counts as a named species at all (as opposed to an ordinary, already-expected density fluctuation, which also has $\mathcal{I} > 0$ in this definition), and the reference scale ℓ, are not fixed by anything stated so far – exactly like Σ's dependence and the Raychaudhuri source term, this is a free choice the hypotheses need to make explicitly before $\mathcal{I}$ does predictive work rather than descriptive work.

How this maps onto the taxonomy (§4.12). A natural first pass: *Sphere* = a trapped surface is present, regardless of $\mathcal{I}$; *Vortex* = $\mathcal{I} > 0$, $\omega_{\mu\nu}$-dominated, no trapped surface; *Depression* = $\mathcal{I} > 0$, small and localized, $\theta < 0$-dominated, no trapped surface; *Crack/channel* = $\mathcal{I} > 0$, $\sigma_{\mu\nu}$-dominated, elongated, no trapped surface. This is offered as a starting point for filling in the taxonomy table's open columns, not as a settled result.

4 Hypotheses

4.1 Hypothesis 1: Gravity as Elasticity of Space

Statement (scoped form). Space can bend, and space has elasticity. Gravity – as a measurable quantity, a value of g, a redshift, an orbit – is calculable from the amount of bend, through a fixed elastic constant of space. H1 takes no position on why bending happens, or why mass responds to it; that causal question is left entirely to H2. H1's claim is quantitative, not mechanistic: how much bend, converted to how much gravity, via a constant that does not vary from body to body.

Defensible core (established; checked directly against five bodies). Three specific points survive direct numerical checking and are adopted as this hypothesis's settled foundation:

(1) Elasticity is a universal constant, not a variable one. Defining $\Sigma(x) := c^4/(8\pi G(x))$ (§3.2), the question of whether Σ varies from body to body reduces to the question of whether G does. It does not, to the precision of every available test: G is measured identically for Earth, the Sun, Mars, Jupiter, and Saturn (recovering their actual surface gravities via $g = \frac{4}{3}\pi G\rho R$ to the percent level once each body's own ρ and R are used), and independent bounds on $\dot{G}/G$ from lunar laser ranging and pulsar timing constrain any time variation to parts in 10^{11}–10^{12} per year. Σ is adopted here as Σ_0, constant, on this evidence – not as an unexamined assumption.

(2) The amount of bend is set by mass and distance from the body's center, not by density and radius taken as independent ingredients. The more fundamental law, by Newton's shell theorem, is that the field at a point depends only on the enclosed mass M and the distance r from the center: $g(r) = GM/r^2$ outside any spherically symmetric body, regardless of

density or internal structure. "Density and radius" is the special case of this law evaluated at a body's own surface ($r = R$), where $M = \rho \cdot \frac{4}{3}\pi R^3$ folds density and radius together; it is not a separate, additional dependence. Checked directly: a body with Earth's mass but twice Earth's radius (density $\approx 689\,\mathrm{kg/m^3}$) gives the same g at that radius as real Earth gives at the same distance from its own center – confirming density itself carries no independent information once mass and distance are fixed.

(3) H1 computes gravity from bend; it does not assert that bend is gravity's cause. This mirrors the Einstein field equations' own epistemic status: $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}/c^4$ is a correspondence between curvature and mass-energy, not a mechanistic derivation of why mass-energy curves anything – GR has never answered that "why" either. Scoping H1 the same way removes a standing tension with H2: H1 is the quantitative layer (how much bend, how much gravity, and the fixed conversion constant between them); H2 is the mechanistic layer (why bending happens, why mass "sinks"). They stack rather than compete for the same explanatory job.

A taxonomy of bending, and gravity as a journey into slower time. This paper's own material gives exactly three distinct ways something can bend space, worth stating as a set rather than leaving scattered: (1) a mass simply sitting in space – the passive, source-centered case H1 itself describes, with no additional process invoked, elasticity alone setting the amount; (2) a *centripetal* process – something continuously sinking toward a center (H2a's candidate driver) – bending space further via the equivalence principle's acceleration-as-field correspondence; (3) a *centrifugal* process – something flung outward from an axis by rotation (H2b's candidate driver) – the only other acceleration-based route that correspondence makes available. Items (2) and (3) are H2's proposed causal layer, not H1's; H1 by itself takes no position on which, if either, is at work.

Phrased in terms of the mechanism already derived in §4.3 – that the acceleration a falling body feels is exactly c^2 times the spatial gradient of its own clock rate, $a = -\partial_r \Phi$ – gravity admits a vivid, concrete restatement: *falling is a journey toward where time passes more slowly.* Relative to a distant clock, the falling body's clock increasingly lags, falling further behind the longer the journey continues – loosely, a journey "back" in time in the sense that the gap between its own elapsed time and the distant clock's grows without bound. This is a restatement of the g_{00}-gradient mechanism already established, not a literal reversal of time's direction: nothing in this journey runs backward, it only ever runs slower. It is also not a new claim distinct from H8's own "feature of time" language (§4.9) – it is the same mechanism, stated here at the level of ordinary gravity rather than at the extreme, core-sinking limit H8 describes.

The deepening of this lag with bend, checked numerically. The lag rate at radius r is $\Phi(r)/c^2 = -GM/(rc^2)$ – "how much bend" in the time-lag sense – and it deepens as r decreases, exactly tracking the acceleration, since $a = -\partial_r \Phi$ is the lag rate's own spatial slope. For Earth: at the surface ($r = 6.371 \times 10^6\,\mathrm{m}$), the lag rate is $\approx -6.95 \times 10^{-10}$; at GPS orbit ($r = 2.657 \times 10^7\,\mathrm{m}$), it is $\approx -1.67 \times 10^{-10}$ – about four times shallower. Their difference, over one day, reproduces the measured GPS correction directly:

$$(6.95 - 1.67) \times 10^{-10} \times 86{,}400\,\mathrm{s} \;\approx\; 4.56 \times 10^{-5}\,\mathrm{s} \;=\; 45.6\,\mu\mathrm{s/day}, \tag{4}$$

matching §5.3's already-cited 45.7 μs/day figure. Over the same two altitudes, g itself falls from $9.82\,\mathrm{m/s^2}$ at the surface to $0.565\,\mathrm{m/s^2}$ at GPS orbit – both the lag and the acceleration deepen together with the bend, as they must, since one is literally the spatial slope of the other.

A second layer worth naming: the gaps between successive steps of this fall are not constant either. $a = -\partial_r \Phi$ is itself a function of r, not a fixed value, so as a body falls closer in, the *increments* of lag and speed it picks up per step of distance fallen themselves grow – a second derivative, not just a first:

$$\frac{d^2\Phi}{dr^2} = \frac{2GM}{r^3}, \tag{5}$$

which is exactly the tidal/curvature term already used in §5.1 and §4.3's discussion of true, frame-independent curvature (for Earth, $\approx 3.1 \times 10^{-6}\,\mathrm{m/s^2}$ per meter at the surface, ≈ 3086 Eötvös) – the one quantity that stays nonzero in every frame, unlike a centrifuge's merely apparent pseudo-gravity. This is worth flagging as a genuinely open thread rather than a settled result: whether gravity's full story is better told through this second-derivative, widening-gap structure, rather than through the first-derivative acceleration term alone, is not resolved here – the identification above only establishes that the familiar tidal term already in this paper is the natural home for that intuition, not that it supplies a new mechanism beyond what tidal curvature already does in standard GR.

A note on scope, restated for this point specifically. None of this is offered as a claim that GR is wrong. GR's agreement with experiment – Mercury's perihelion, light bending, binary-pulsar orbital decay, direct gravitational-wave detections, the GPS figure re-derived above – is not in question. The claim worth taking seriously is narrower: GR's two standing simplifying assumptions, that spacetime is smooth and that empty space contributes nothing of its own, are assumptions, not something GR itself proves, and this paper's own working postulates (P1–P3) are an attempt to ask what, if anything, changes if one or both are relaxed – while requiring, as H1 and H1b already do, that any such change vanish exactly in every regime where GR has already been tested to the precision above. **Formal statement.**

$$g(r) \;=\; \frac{GM}{r^2} \;=\; \frac{c^4}{8\pi\Sigma_0}\cdot\frac{M}{r^2}, \qquad \Sigma_0 = \frac{c^4}{8\pi G} \approx 4.98\times10^{42}\,\mathrm{N}\ \text{(constant)}. \tag{6}$$

Compare to the Einstein field equations, which already relate curvature to mass-energy through this same fixed coupling:

$$G_{\mu\nu} \;=\; \frac{1}{\Sigma_0}\,T_{\mu\nu}. \tag{7}$$

In this scoped form, H1's content is exactly the recovery of standard Newtonian/GR surface gravity, with Σ_0 playing G's role – no more, no less, until a variation in Σ is proposed (see H1b, below).

Motivation. Gravity in general relativity already states that curvature is produced by mass-energy, through the constant G. H1 reinterprets that constant as a physical elasticity – a relabeling with a clear, checkable meaning, not yet an independent discovery, because every number checked so far is the one standard physics already predicts.

What would falsify it / what is still undefined. Points (1)–(3) above are no longer open items: they have been checked directly and incorporated as this hypothesis's settled core. What remains open is whether H1 has anything to say *beyond* this recovery of standard gravity: (i) As stated, with Σ held constant, H1 makes no prediction that differs from standard GR and is therefore not yet falsifiable in its own right – it becomes a hypothesis with independent content only if Σ is allowed to vary (with density, curvature, or the irregularity scalar $\mathcal{I}$, §3.2) in a way the measured constant G does not, and only in a regime not already excluded by the $\dot{G}/G$ bounds in point (1). (ii) Any such variation must still reduce to the constant Σ_0 everywhere G has actually been measured. (iii) A genuinely novel extension of H1 (provisionally, H1b) would propose a specific, additional contribution to bend tied to $\mathcal{I}(x)$ – e.g. $g_{\text{total}} = GM/r^2 + f(\mathcal{I})$ – with a stated magnitude for f, checkable against regions of high turbulence or inconsistency (H3–H6); this is flagged as the open direction for future work, not yet developed here.

4.2 Hypothesis 1b: Variable Elasticity Tied to Local Field Conditions

Statement (rough form). H1's constant Σ_0 was checked only against *which body* is sourcing the field (Earth vs. the Sun vs. Jupiter, etc.) and found not to vary that way. H1b asks a different question: does Σ depend on *local field conditions* – how much space is already bent or disturbed right where the field is being evaluated – rather than on which body is doing the bending? If so, the exact shape-indifference that follows from constant Σ (any mass distribution with the same total M produces the same external field, §4.1 point 2) would no longer hold exactly: a tightly concentrated mass and a spread-out mass of the same M could, in principle, leave measurably different fields outside themselves, because the intervening space's own elasticity would no longer be a neutral, fixed background.

Formal statement. Generalize Σ from a constant to a field that responds to the irregularity scalar $\mathcal{I}(x)$ already defined in §3.2:

$$\Sigma(x) \;=\; \Sigma_0\Big[\,1 + \lambda\, h\big(\mathcal{I}(x)\big)\Big], \qquad h(0) = 0, \tag{8}$$

where λ is a dimensionless coupling (sign and magnitude unspecified) and h is some function, also unspecified, vanishing when space is locally undisturbed ($\mathcal{I} = 0$, recovering $\Sigma = \Sigma_0$ exactly). The field equation becomes $G_{\mu\nu} = T_{\mu\nu}/\Sigma(x)$, now genuinely nonlinear, since $\mathcal{I}(x)$ is itself built from the kinematic quantities $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$ that the curvature equation also governs.

Motivation. This is the only way, within this paper's own formalism, to recover the shape-sensitive intuition the honey/ping- pong-ball analogy was reaching for (§4.1 discussion): mechanical media (honey) respond to an object's shape because they interact through real contact forces; a field theory with constant Σ cannot, by the shell theorem, however counterintuitive that is. Letting Σ respond to local

field conditions – rather than to which body is present – is the one lever left that could restore shape-sensitivity without contradicting anything already confirmed about Σ not differing between bodies.

A motivating image: pseudo-centripetal coupling. Picture a lift (space) rising at a perfectly steady speed, with a vertical bar (time) passing through it, fixed to the outside world. Letting the bar slide loosely through your grip costs nothing; gripping it tighter resists the slide, and that resistance is what presses you into the floor – the felt, outward, normal-force side of weight (§3.2's distinction between felt and free-fall acceleration), not the inward free-fall motion itself, hence *pseudo*-centripetal. The image's only job here is to motivate why a coupling between a body and the local flow of time might vary at all: a looser or tighter grip is a direct stand-in for a smaller or larger $\lambda\, h(\mathcal{I}(x))$ in the formal statement above.

Where this agrees with established physics, and where it doesn't. It agrees, and is simply a restatement, wherever it reduces to the g_{00}-gradient mechanism already derived and numerically checked in §4.1/§4.3 (the GPS figure, the $9.82 \to 0.565\,\mathrm{m/s^2}$ comparison): that mechanism does not depend on this image at all, and holds regardless of whether the image is accepted. Where it departs is the specific claim this image is actually standing in for – that the *strength* of that coupling (Σ, the "grip") is itself variable rather than the fixed Σ_0 every existing test has measured. That departure is not a restatement of anything in GR; it is H1b's own open claim, tiered (c) exactly as stated above, and it inherits every open problem already listed below – most directly, that h and λ are unspecified, and that wherever Σ_0 has already been confirmed constant (the solar system, laboratory G, pulsar timing), this coupling must be shown to go to zero there, not simply asserted to. The honest way to use this hypothesis, per its own stated purpose, is as a question still being asked – why a body accelerates at all, one level beneath where GR's own geodesic equation stops explaining and starts simply describing – not as a settled answer competing with GR's tested results.

Relation to existing programs. This is not a new category of idea – it is close to two existing, well-developed, and tightly constrained research programs. **Scalar-tensor gravity** (Brans–Dicke, 1961, and successors) promotes G itself to a dynamical scalar field, $1/G \propto \phi(x)$, sourced by its own field equation – formally almost exactly H1b's $\Sigma(x)$, just coupled to the trace of $T_{\mu\nu}$ directly rather than to $\mathcal{I}(x)$. Solar-system tests already constrain this precisely: Cassini-spacecraft Shapiro time-delay tracking (Bertotti, Iess & Tortora, 2003) bounds the Brans–Dicke parameter $\omega > 40{,}000$, meaning any scalar-tensor-style deviation from constant G in the solar system is constrained to be extremely small – a direct, numerical precedent for how tightly H1b's $\lambda\, h(\mathcal{I})$ term is already bounded wherever $\mathcal{I}$ is solar-system-sized. **MOND** (Milgrom, 1983; already discussed for H6, §2.3) is the nearer precedent for the *style* of H1b's claim – a response that activates based on a local field condition rather than on source identity – though MOND's trigger is a universal acceleration scale a_0, not $\mathcal{I}(x)$, and the two should not be treated as the same proposal without showing they coincide.

What would falsify it / open problems. (i) $h(\mathcal{I})$ is not yet specified; until it is, H1b makes no prediction and is not yet falsifiable. (ii) Wherever Σ has already been confirmed constant – the solar system, laboratory G measurements, pulsar timing (§4.1 point 1) – $\mathcal{I}$ is presumably small, but this needs to be checked explicitly: h and λ must be small enough there that $\Sigma \approx \Sigma_0$ to the precision already measured (at minimum, as tightly as the Cassini $\omega > 40{,}000$ bound already requires of comparable theories). (iii) H1b therefore only has content if a regime exists where $\mathcal{I}$ is large enough for $\lambda\, h(\mathcal{I})$ to produce a checkable deviation – candidates include near the compact, high-curvature cores invoked by H3/H4/H8, or cosmic-web filaments (H3/H4's taxonomy) – and a specific prediction in one such regime, not yet stated here, is the next concrete step. (iv) Distinguishing H1b's $\mathcal{I}$-dependence from MOND's a_0-dependence, and from plain scalar-tensor gravity's direct $T_{\mu\nu}$-coupling, is necessary before H1b can claim to be a distinct proposal rather than a relabeling of either. (v) *Birkhoff's theorem is the specific obstacle, and evading it should be named for what it is.* A static, spherically symmetric vacuum exterior is forced to be exactly Schwarzschild whenever the coupling is constant (§5.1). A Σ that responds to $\mathcal{I}$ can only evade that conclusion by putting real stress-energy, or a genuine additional field, into what GR would otherwise treat as empty exterior space – which is an amendment to the Einstein field equations, not a parameter varying within them. H1b should state this amendment explicitly, rather than let "Σ varies" stand in for a change of theory it has not yet written down as one.

H1b, extended: a single scalar Medium/Field unifying H1b, H6, and H7. Item (i) above is $h(\mathcal{I})$'s missing specification; this is the paper's current candidate for supplying it, by proposing that $\mathcal{I}(x)$ is not merely a bookkeeping scalar built from the kinematic triplet, but the local value of a genuine dynamical field, $\mathcal{I}(x) \to \phi(x)$, with its own action and its own equation of motion – the same field this paper has so far discussed, piecemeal, as the energy behind cosmic expansion (H7) and as the source of local disturbances (H3/H5/H6's taxonomy). Writing one field to do all three jobs, rather than three

separate unspecified ingredients, is the point of stating it explicitly here.

Proposed action.

$$S = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\big(1+\lambda\,h(\phi)\big)R - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi)\right] + S_{\text{matter}}[g_{\mu\nu},\psi]. \tag{9}$$

The non-minimal coupling $f(\phi) = 1 + \lambda h(\phi)$ reproduces H1b's $\Sigma(x) = \Sigma_0\big[1 + \lambda h(\mathcal{I}(x))\big)\big]$ exactly, now with $\mathcal{I}(x) \equiv \phi(x)$ rather than a separately-defined kinematic scalar – H1b's open coupling and this field's non-minimal term are the same object, not two. Varying with respect to $g^{\mu\nu}$ and ϕ gives, schematically, a modified Einstein equation sourced by both ordinary matter and ϕ's own stress-energy, and a Klein–Gordon-type equation for ϕ with an extra source term from $f'(\phi)R$ (the standard scalar-tensor structure; full derivation and the precise $f'(\phi)R$ back-reaction term are open work, not carried out here).

The field's two roles. Split $\phi = \bar{\phi}(t) + \delta\phi(x)$.

- **Background, $\bar{\phi}(t)$ – H7's role.** For a slowly rolling field with $V(\bar{\phi})$ dominating over $\dot{\bar{\phi}}^2$, $p_{\bar{\phi}} \approx -\rho_{\bar{\phi}}$: the field behaves like dark energy, supplying H7's expansion-energy smoothly and uniformly, with no disturbance anywhere.
- **Perturbation, $\delta\phi(x)$ – H3/H5/H6's role.** Departures from the smooth background source local curvature directly, through the same field equation, and simultaneously modulate Σ locally through $f(\phi)$ – one departure from uniformity doing both jobs H1b and H3/H5/H6 each needed their own unspecified ingredient for.

What this buys, concretely. H1b's $h(\mathcal{I})$ stops being an arbitrary placeholder and becomes $h(\phi)$, a function of a field with its own dynamics – falsifiable in principle once $V(\phi)$ and $f(\phi)$ are chosen, rather than unconstrained by construction. H3/H5/H6's "inconsistencies" and "vortices" become a single, named object – $\delta\phi(x)$ – rather than a family of separately-invented mechanisms. And the paper's P1–P3 postulates stop being three independent claims about a medium and become one concrete field content for that medium.

What this does not yet buy, stated as plainly as every other open item in this paper. (1) The Cassini bound, $\omega > 40{,}000$, inherited already by H1b, now constrains $f'(\phi)$ directly wherever the solar system sits on ϕ's potential – this must be checked once $V(\phi)$ and $f(\phi)$ are specified, not assumed satisfied. (2) $\delta\phi$'s required energy density, as a function of galactic radius, still has to reproduce the same $M(r) \propto r$ profile dark matter currently supplies for flat rotation curves – unifying the *ontology* behind H6 and H7 does not reduce the *quantity* of structured energy H6 needs; that budget is still fully owed. (3) This is not yet a novel claim: unifying dark energy and the dark-matter role into one scalar field is an existing, actively studied category (unified dark fluid / Chaplygin-gas models; clustering quintessence; scalar-field and BEC dark matter), and none of these has yet been shown, in the literature, to reproduce galaxy-by-galaxy rotation curves with the same field also driving cosmic acceleration – this paper inherits that open problem rather than solving it by naming ϕ. (4) $V(\phi)$ and $f(\phi)$ remain completely unspecified; choosing them, and checking the resulting $\bar{\phi}$ background against the measured dark-energy equation of state ($w \approx -1$, within current observational bounds), is the concrete next step this extension is building toward, not a result already in hand.

4.3 Hypothesis 2: Gravity as the Felt Consequence of Space-time Curvature

Statement (rough form). Gravity is, at root, either a centripetal process (something continuously sinking toward a center) or a centrifugal one (something flung outward from an axis) – and the two make opposite, checkable predictions for the direction of gravitational time dilation. A clock held near a mass runs slower than a distant clock (confirmed daily by the GPS correction and directly by optical-clock height comparisons): exactly the signature of a centripetal process, where the body being accelerated is pulled toward the "floor" of an accelerating frame, not a centrifugal one, which would require the opposite sign (§4.3's thought experiment below works through this in full). This favors a centripetal origin for gravity over a centrifugal one before any of the rest of this subsection's objections are even reached. (A precision worth stating once: the acceleration an accelerometer actually *feels* standing on Earth's surface points outward – the normal force balancing gravity – while it is the *free-fall* acceleration of an unsupported body that is centripetal, i.e. center-seeking. "Centripetal" here describes that free-fall tendency, the thing H2 is trying to source, not the felt support force.)

Revision, superseding the paragraph above's original framing. This subsection originally proposed the reverse order – bent space as a *consequence* of some other driver (cosmic expansion or spin), with gravity itself caused by that driver rather than by the bend. Having checked both candidate drivers against real numbers, this paper now adopts the opposite, and simpler, order as its defining statement: spacetime curvature, sourced by mass, is the cause; the felt force called gravity is its consequence. (2a) is retired as a *driver* for exactly this reason – there is no need for cosmic expansion or any other external driver once curvature itself is allowed to be the cause – and is kept only as the label for the centripetal, curvature-is-the-cause reading developed below. (2b) is kept as the centrifugal candidate specifically because it lets this paper show, concretely, what gravity is *not*: a spin-based felt force with the same sign as real gravity but zero true, frame-independent curvature behind it (§4.3's discussion of the tidal/Riemann distinction). By the equivalence principle, a sustained felt acceleration of $9.81\,\mathrm{m/s^2}$ (an elevator, a rocket) is indistinguishable, locally, from a gravitational field – but only the radial, curvature-sourced case (labeled "centripetal" here only as shorthand, not in the strict textbook sense; see the formal statement and its note just below) is genuine gravity under this paper's definition; the centrifugal case merely mimics it.

Formal statement, current. Gravity, under this paper's defining statement, is

$$a^i \;=\; -\partial_i \Phi \;=\; -c^2 \Gamma^i_{00}, \tag{10}$$

the Newtonian limit of the geodesic equation, built entirely from $g_{00} \approx -(1 + 2\Phi/c^2)$ with no spatial-metric term entering at this order (§4.3's H2c discussion below). This is the equation this paper stands on for H2; the two boxed equations below are retired and kept only as a record of the hypothesis's original, superseded form.

Derivation: from the clock-rate gradient to $a = -\partial_r \Phi$. Thought Experiment 1 below pictures gravity as the widening gap between neighboring clock *rates*; this derivation shows that picture is not a gloss laid over the formal statement above but the same quantity, differentiated. Let $d\tau/dt$ denote a static clock's proper-time rate at position r, relative to a distant reference clock. In the weak-field, static limit,

$$\frac{d\tau}{dt} = \sqrt{-g_{00}} \approx \sqrt{1 + 2\Phi/c^2} \approx 1 + \frac{\Phi}{c^2},$$

so $d\tau/dt$ *is* the local clock-rate factor Thought Experiment 1 calls "how tightly you grip the bar." Differentiating this factor with respect to r and rescaling by $-c^2$ gives

$$-c^2\, \partial_r \left(\frac{d\tau}{dt} \right) = -\partial_r \Phi = \frac{GM}{r^2} \qquad (\Phi = -GM/r),$$

exactly Newtonian gravitational acceleration, obtained purely from how fast the clock-rate gradient changes with position, with no spatial-metric term anywhere in the calculation. This is not an independent check of the formal statement above; it is the same object, $a^i = -c^2\Gamma^i_{00}$, written out explicitly, since for a static, weak-field metric $\Gamma^i_{00} = \frac{1}{2}\partial_i g_{00}$ to leading order, so $a^i = -c^2\Gamma^i_{00} = -\frac{c^2}{2}\partial_i g_{00} = -\partial_i \Phi$ falls out immediately from differentiating g_{00} itself – and $d\tau/dt$ is just $\sqrt{-g_{00}}$, so differentiating the clock-rate factor and differentiating g_{00} are, to this order, the same step twice. The "widening gap between local clock rates" in Thought Experiment 1 is therefore, literally, $-c^2\partial_r(d\tau/dt)$: a physical name for this derivation's own gradient, not a separate mechanism running alongside it.

Formal statement, original (retired).

$$\text{(2a)-engine, withdrawn:}\quad \vec{a}_{\text{gravity}} \stackrel{?}{=} \vec{a}_{\text{cosmic expansion}}(\text{mass, position}),$$
$$\text{(2b), not true gravity:}\quad \vec{a}_{\text{gravity}} \stackrel{?}{=} \omega^2 r\, \hat{\rho} \text{ (centrifugal form)}$$

(2a), revised: gravity as the felt consequence of spacetime curvature itself. (2a)'s original engine – a literal, ongoing acceleration of cosmic expansion, mapped onto gravity via the equivalence principle – is retired as stated, for the reason already derived numerically below (objection (ii)): $cH_0 \approx 6 \times 10^{-10}\,\mathrm{m/s^2}$ is roughly ten orders of magnitude short of $9.81\,\mathrm{m/s^2}$, and is the same everywhere rather than centered on Earth. The problem is specifically this magnitude and shape mismatch, not that a sustained cosmic acceleration is implausible on its own terms (the universe's expansion genuinely is accelerating, and under ΛCDM is expected to keep doing so indefinitely) – it is simply the wrong size and the wrong shape to be Earth's gravity.

What (2a) keeps, having dropped that engine, is the conclusion already established above: gravity is radial and mass-centered – this paper's loose use of "centripetal" for that radial character, not centrifugal,

in character. (A terminological note, since the word is used repeatedly below: strictly, "centripetal" names the sideways acceleration, v^2/r, that holds a body at constant speed on a circular path. The term this paper actually needs – $-\nabla\Phi$, radial, changing a body's speed from rest – is not that. This paper keeps using "centripetal" only as a loose label for "radially inward, toward a mass," and the reader should not read any of the stricter, circular-motion content of the word into it.) What produces that radial character is spacetime curvature itself, sourced by mass, with no separate ingredient standing between them: each point in space is connected, by the curvature there, to a local rate of clock flow slightly slower than its neighbors' (not an earlier *date* or moment – a lower clock ticks slower, it does not sit in the past; rate and date are different things, and nothing in this paper's mechanism picks out a preferred time direction); as the curvature deepens, the gap between neighboring clock *rates* widens, and that widening gap *is* the gravitational acceleration ($a = -\partial_r\Phi$, the Newtonian limit of the geodesic equation, §4.1). What the earlier draft of this section called a "coupling" is this same quantity under a different name, not an additional thing curvature causes – the lift/bar image below should be read as a picture of curvature itself, not of a separate mechanism sitting alongside it.

Thought Experiment 1: the lift and the time bar. A lift (space) rises at a perfectly steady, non-accelerating pace. A vertical bar (time), fixed to the world outside the lift, passes through a slot and slides freely past as the lift rises. Each point along the bar marks a different local clock *rate*, not a different date – farther down the bar, clocks run slower, not earlier. Reaching out and taking hold of the bar is connecting yourself to one such rate:

- A *loose* grip connects you to a rate only slightly slower than the lift's own; the gap is small, and the felt pressure into the floor is slight.
- A *firmer* grip connects you to a slower rate still; the gap is larger, and the felt pressure is correspondingly greater.
- A *fully locked* grip connects you to the rate at the point of deepest curvature, where the gap is largest and the pressure is maximal – squashed to the floor.

How tightly you grip is not an independently chosen dial – it is set by how much curvature is present at your location, exactly as the bend in H1 is set by mass and distance, not chosen freely. Note again what never changes: the lift's own speed. It stays steady throughout; nothing about the lift itself accelerates.

Summary. The felt weight here is produced neither by the lift accelerating (it doesn't) nor by true center-seeking motion as a separate cause – it is produced directly by the curvature's own structure: the widening gap between successive local clock rates as curvature deepens. This gap-widening is the single thing being pictured, under two names – "grip" in the lift image, $a = -\partial_r\Phi$ in the formal statement, and curvature in the geometric language of §3.2. None of these three is prior to, or causes, the others; they are the same fact.

Definition of gravity, revised. Gravity is the felt consequence of spacetime curvature: specifically, the widening gap between the local clock rate at one point in space and the local clock rate at a neighboring point, as the mass-sourced curvature between them deepens. This is a restatement, not a new mechanism – it is the Newtonian limit of the geodesic equation, and it is consistent with the GPS correction, the Chou et al. clock data, and the Earth tidal-gradient figure (§4.1, §4.3's discussion above) – but those three are one potential, $\Phi = -GM/r$ for Earth, read three ways, not three independent confirmations of a mechanism beyond GR. It pins the *mechanism* down as real and already well established, not illusory – but it does not, by itself, answer why mass curves spacetime's time axis this particular way in the first place; that remains exactly as open as it is under plain GR (§4.1, point 3). H1b's proposed variability of $\Sigma(x)$, if it means anything under this reading, has to be understood as curvature having its own local texture ($\mathcal{I}(x)$, already built from curvature-adjacent kinematic quantities) rather than as a second, independently-sourced coupling layered on top of curvature – otherwise H1b would be proposing an ingredient this section has just shown does not exist separately from curvature itself.

Motivation. The equivalence principle is real and well-tested: locally, uniform acceleration and a uniform gravitational field are experimentally indistinguishable. H2 extrapolates this local equivalence into a global claim about the *origin* of every gravitational field.

What would falsify it / open problems. Objections (i)–(v) below all target *H2-engine* – the retired cosmic-expansion or cosmic-spin driver described in the original Statement above – not the curvature-is-cause revision this paper now keeps (H2-revised, §4.3's "(2a), revised" discussion); H2-revised is the standard Newtonian limit of GR's geodesic equation and is not challenged by any of (i)–(v), precisely because it no longer posits any of the drivers these objections are about. (i) Known gravitational fields (Earth's, the Sun's) are centered on specific masses, static to high precision in the lab frame, and fall

off as $1/r^2$ from that mass; the hypothesis as stated does not yet show how a universe-wide expansion rate or a universe-wide spin rate produces a field with exactly that mass-centered, inverse-square structure rather than a uniform or axis-centered one. (ii) The measured acceleration of cosmic expansion (from dark energy) is many orders of magnitude too small, and far too uniform/non-local, to supply $9.81\,\mathrm{m/s^2}$ at Earth's surface without an additional, yet-unstated amplification mechanism tied to local mass – concretely, the velocity-drift acceleration $cH_0 \sim 3 \times 10^8 \times 2.18 \times 10^{-18} \approx 6 \times 10^{-10}\,\mathrm{m/s^2}$ is roughly ten orders of magnitude short of Earth's g, and the fractional rate $\ddot{a}/a \sim H_0^2 \sim 4.75 \times 10^{-36}\,\mathrm{s^{-2}}$ is not itself an acceleration (its units are $\mathrm{s^{-2}}$, not $\mathrm{m/s^2}$) and so cannot be compared to g directly; converting it to an acceleration requires multiplying by a length scale, and the natural one here is Earth's own radius: $H_0^2 R \approx 4.75 \times 10^{-36} \times 6.371 \times 10^6 \approx 3 \times 10^{-29}\,\mathrm{m/s^2}$, roughly *thirty* orders of magnitude short of g, not ten. Either way the shortfall is decisive, and neither quantity is centered on the Earth to begin with. (iii) A *global* rotation of the universe – a single, coherent spin axis for the universe as a whole – is tightly constrained by the observed isotropy of the cosmic microwave background (§2.1) to be far too small to source planetary-scale gravity. This bound is specifically about a global, uniform rotation; it says nothing about an undefined *local* spin, so (2b) needs to say explicitly whether "spin of space" means this tightly-bounded cosmic rotation or a different, as yet undefined, local spin – the two are not interchangeable, and only the former currently has a precise observational bound attached. (iv) Both variants currently need the equivalence principle to do more work than it is usually read as doing: it licenses treating acceleration *as* a gravitational field locally, but does not by itself explain why mass generates an inverse-square field centered on itself. (v) *Uniform driver, mass-proportional response.* The posited driver – cosmic expansion or cosmic spin – is, by hypothesis, the same everywhere; it does not vary from body to body. Yet the resulting bend is observed to scale with mass ($g \propto M$). A uniform acceleration applied equally everywhere does not, by itself, explain why a more massive body sinks or bends space *more* than a less massive one – getting that mass-proportional shape out of H2's story requires a second, separate assumption about how mass modulates the response to the driver, and that assumption is, structurally, H1's mass-coupling (Σ) again, reintroduced rather than derived from the acceleration/spin mechanism itself. H2 therefore does not yet avoid needing H1's coupling; it relocates where that coupling has to be assumed, rather than explaining it away. Pressing this is also what clarifies the alternative, non-circular reading available to H1 on its own (§4.1, point 3): that mass-space coupling may simply be a brute, unexplained interaction – exactly the epistemic status the Einstein field equations themselves already have, with no deeper derivation offered by GR, string theory, loop quantum gravity, or any other current program either.

Thought Experiment 2: distinguishing centripetal (2a) from centrifugal (2b) by the sign of time dilation. The equivalence principle makes a specific, checkable prediction for each variant, and the two disagree with each other, which gives a reader a concrete way to prefer one over the other independent of §4's other objections.

Under (2a), constant acceleration (the elevator/rocket case): a clock near the "floor" – the end of the elevator in the direction of the acceleration, analogous to Earth's surface – runs *slower* than a clock near the "ceiling," exactly matching the measured fact that clocks on Earth's surface run slower than clocks farther from Earth (confirmed daily via the GPS correction, and directly via optical-clock experiments over height differences of a few centimeters). This is not a new derivation; it is the same acceleration-based time dilation already used, uncontroversially, to explain why the equivalence principle holds for clocks as well as for felt force.

Under (2b), constant spin (the rotating-drum case): the relevant comparison is between a point near the rotation axis and a point farther out, at the radius standing in for Earth's surface. Here it matters precisely *how* a body is supposed to be moving with the spin. If a body is carried along *with* the rotating medium, with no motion relative to its own local surroundings (the reading (2b) needs, since it treats the spin as a property of space itself rather than of bodies moving through static space), no established result in general relativity makes such a body's clock run faster than distant, non-rotating space – comoving observers in an expanding universe and frame-dragged observers near a real rotating mass both show the opposite pattern, or no net effect from the carrying alone. If instead the body is read as kinematically moving through otherwise-static space at the rotation rate (ordinary special-relativistic motion, $v = \omega r$), that body's clock runs *slower* farther from the axis – the right sign, but for a reason that only applies to a body genuinely moving relative to space, which is a different, narrower claim than (2b) as stated ("spin of space" itself, not bodies moving within static space).

Conclusion a reader can draw: (2a) reproduces the measured sign of Earth's gravitational time dilation directly and without qualification. (2b) reproduces it only by quietly switching to the kinematic, bodies-moving-through-space reading, which is not the claim (2b) was making; under the reading (2b)

actually needs – space itself carrying bodies along – no established mechanism supplies the correct sign at all. This gives a concrete reason to prefer (2a) over (2b) specifically, independent of (and in addition to) objections (i)–(iv) above. It does not rescue (2a) from those other objections – the magnitude of cosmic expansion's acceleration and the mass-centered shape problem in (i)–(ii) are untouched by this result and remain fully open.

H2c (bend is geometry, not force). The spatial curvature H1 computes redirects a moving body's *direction* without changing its *speed*, and does so independent of how fast the body is moving. This is a real effect, not an invented one – it is exactly the velocity-independent term that supplies half of gravitational light-bending ($\delta_{\text{spatial}} = 2GM/bc^2$, against a total $\delta = 4GM/bc^2$ for light). But it is not, by itself, an accelerating force, and should not be mistaken for the whole of gravity.

Formal statement. For a body of speed v passing a mass M at impact parameter b, the total small-angle deflection splits as

$$\delta \approx \underbrace{\frac{2GM}{bv^2}}_{\text{Newtonian } (g_{00}) \text{ term}} + \underbrace{\frac{2GM}{bc^2}}_{\text{spatial-curvature term, independent of } v}, \tag{11}$$

with ratio $\delta_{\text{spatial}}/\delta_{\text{Newton}} = v^2/c^2$. The first term is the ordinary Newtonian deflection from $-\nabla\Phi$ (equivalently, from g_{00}); it is *not* centripetal in the strict sense – it is radial and changes the body's speed, where centripetal acceleration proper is the sideways, speed-preserving term that holds a body on a circular path. This paper labels it "centripetal" elsewhere only as shorthand for "radially inward," and that shorthand is dropped here, in the one equation in this subsection, to keep the two senses from being conflated.

Numeric check against the measured solar deflection. The total $\delta = 4GM/(bc^2)$ is the standard GR light-bending formula, transcribed here, not independently derived from Σ or from this paper's own machinery; evaluating it is a check that this paper's bookkeeping reproduces a known number, not a new prediction. For light grazing the Sun's limb, $GM_\odot \approx 1.327 \times 10^{20}\,\text{m}^3/\text{s}^2$ and $b \approx R_\odot \approx 6.96 \times 10^8\,\text{m}$, giving $\delta = 4GM_\odot/(bc^2) \approx 8.47 \times 10^{-6}\,\text{rad} \approx 1.75''$ – matching the historically measured solar deflection (Eddington et al., 1919, and all subsequent measurements) to the precision quoted. This specific number was not checked in §5 before this revision; light bending is accordingly removed from that section's list of not-yet- checked constraints, having now been evaluated. Mercury's perihelion precession, next on that same list, is checked immediately below.

Numeric check against Mercury's perihelion precession. The same g_{00}-sourced mechanism H2a relies on, combined with H1's spatial-curvature term, gives the standard GR per-orbit perihelion advance for a body on an elliptical orbit of semi-major axis a and eccentricity e,

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}, \tag{12}$$

transcribed here exactly as with the light-bending formula above – a check of this paper's bookkeeping, not an independent derivation. For Mercury, $GM_\odot \approx 1.327 \times 10^{20}\,\text{m}^3/\text{s}^2$, $a \approx 5.7909 \times 10^{10}\,\text{m}$, $e \approx 0.20563$, and orbital period $T \approx 87.969$ days, giving $\Delta\phi \approx 5.020 \times 10^{-7}\,\text{rad} \approx 0.1035''$ per orbit. Over one century (≈ 415.2 orbits), this accumulates to $\approx 42.99''$/century – matching the measured excess precession of Mercury's perihelion, $42.98''$/century (after subtracting all other planets' Newtonian perturbations; Shapiro et al.'s radar-ranging determinations, and the high-precision MESSENGER- tracking value of $42.9799 \pm 0.0009''$/century, Park et al., 2017), to within the precision quoted. Mercury's perihelion precession is accordingly removed from §5's list of not-yet-checked constraints; gravitational-wave speed/waveform constraints remain the one item still outstanding there.

Motivation. §4.1's own defensible-core point 3 already notes that H1 computes gravity *from* the bend without asserting the bend *causes* gravity – a disclaimer, not a proof. H2c supplies the proof: it names the specific sense in which bend alone is insufficient, and ties the missing piece directly to H2a's centripetal/time-dilation mechanism rather than leaving it unaddressed.

What would falsify it / open problems. (i) *A body released from rest has nothing to redirect – and the named mechanism is the resolution, not a gap.* Bend-as-pure-redirection changes the direction of an existing velocity; it has no mechanism for a body with $v = 0$ to begin moving at all. The resolution is that no body is ever actually at rest in 4D spacetime: every object's 4-velocity has fixed magnitude c, so a body "at rest" in space is moving entirely along the time axis. In the weak-field, slow-motion limit, the geodesic equation gives

$$a^i = -c^2\,\Gamma^i_{00} = -\frac{c^2}{2}\,\partial_i g_{00} = -\partial_i\Phi, \tag{13}$$

with $g_{00} \approx -(1 + 2\Phi/c^2)$ – i.e. the spatial gradient of the local clock rate, not the spatial-curvature term this hypothesis isolates, is exactly what bends that time-axis motion into spatial motion, starting from $v = 0$. This is the standard Newtonian-limit derivation of GR's geodesic equation, not a new postulate; it is cited here because it is precisely the mechanism objection (i) shows is missing from bend-as-pure-redirection alone, and it names H2a's centripetal/time-dilation term concretely rather than leaving it as an unspecified placeholder. Every observed instance of a falling object – starting at rest and gaining speed – is this g_{00}-driven acceleration from zero, which the spatial-bend mechanism alone cannot produce. *Checked numerically, against Earth:* the clock-rate field $\Phi(r)/c^2 = -GM/(rc^2)$ deepens from $\approx -1.67 \times 10^{-10}$ at GPS orbit ($r = 2.657 \times 10^7$ m) to $\approx -6.95 \times 10^{-10}$ at the surface ($r = 6.371 \times 10^6$ m); the difference, accumulated over a day, is $\approx 45.6\,\mu$s – matching the measured GPS correction (§5.3) directly. Over the same interval g rises from 0.565 to 9.82 m/s^2: the acceleration is exactly this field's own spatial slope, so the deepening clock lag and the growing acceleration are the same fact, not two correlated ones. (ii) *Even for a moving body, the effect is too small by seventeen orders of magnitude at ordinary speeds.* For $v = 2$ m/s, $v^2/c^2 = (2)^2/(3 \times 10^8)^2 \approx 4.4 \times 10^{-17}$: if gravity were only the velocity-independent spatial-bend term, a slow-moving body would be deflected about 10^{17} times more weakly than measured gravity actually deflects it. (iii) The spatial-curvature term only becomes comparable to the centripetal term as $v \to c$, which is precisely why it is observable for light (gravitational lensing) and unobservable for everyday motion – consistent with, and not a competitor to, H2a's claim that the accelerating mechanism is the time-dilation/centripetal piece, not the spatial bend. (iv) This does not retire H1 – H1's own statement was already scoped to avoid this exact claim (§4.1, point 3) – it closes the loop on *why* that scoping was necessary, and gives H2a's centripetal framing a concrete, numeric reason to be preferred over treating bend itself as the source of force.

4.4 Hypothesis 3: Large-Scale Inconsistencies (Tears) in Space as the Origin of the Largest Black Holes

Statement (rough form). Space is not smooth; it is riddled with inconsistencies in its fabric. Black holes are one type of such inconsistency. Stellar-core-collapse black holes cannot grow very massive. The largest black holes instead arise directly from large tears or gaps in space, produced by uneven cosmic expansion and turbulence.

Formal statement. [PLACEHOLDER: define "inconsistency" as a geometric object – a curvature singularity, a topological defect, a region where the manifold is non-differentiable – before this can be written formally]

Motivation. Correctly notes a real, open astrophysical puzzle: stellar-collapse black holes are bounded in mass (by progenitor mass and pair-instability effects, roughly tens to a few hundred solar masses), yet supermassive black holes (10^6–$10^{10}\,M_\odot$) are observed already in place within the first billion years after the Big Bang – the "seed problem" for supermassive black holes, a genuinely unsolved question in current astrophysics.

What would falsify it / open problems. (i) In standard GR, spacetime is a smooth (differentiable) manifold by construction; a literal "tear" is a topology change, which is classically forbidden and, even in candidate quantum-gravity treatments, is a highly specialized, unresolved research question – not an established mechanism this hypothesis can yet borrow. (ii) The nearest legitimate existing idea – that spacetime is not perfectly smooth at the Planck scale ("quantum foam," Wheeler 1955) – operates at $\sim 10^{-35}$ m, some 30+ orders of magnitude below anything that could seed a supermassive black hole; no established theory currently bridges that scale gap, so it is not yet available as support. (iii) Mainstream astrophysics already has several candidate answers to the seed problem this hypothesis is competing with – direct-collapse black holes from massive gas clouds, heavy seeds from dense early star clusters, and super-Eddington accretion – and H3 needs to say what it explains that these do not, or what observation would distinguish "a tear in space" from a direct-collapse seed.

4.5 Hypothesis 4: Widespread Small Inconsistencies as the Seed of All Accretion

Statement (rough form). Inconsistencies in space are common and widespread, not just the rare large ones behind H3. These smaller inconsistencies are small tears or holes that trap particles, seeding the accretion of matter into stars, planets, and asteroids. Accretion itself then bends space further,

accelerating further collection. It would not be wrong to say a black hole lies at the center of every body in space.

Formal statement. [PLACEHOLDER: as with H3, "trap" and "hole" need a precise geometric or field-theoretic definition before this is an equation rather than a picture]

Motivation. Correctly identifies that accretion is how stars, planets, and asteroids form, and that growing mass deepens a gravitational well, which is consistent with standard accretion theory (gravitational instability and core-accretion models for star and planet formation). The closest existing idea to "a small hole seeding accretion" is the primordial black hole (PBH) hypothesis (Zel'dovich & Novikov, 1966; Hawking, 1971), in which density fluctuations in the very early universe collapse directly into small black holes, which have since been proposed as possible seeds for structure formation and as a dark-matter candidate.

What would falsify it / open problems. (i) Standard star and planet formation is already well modeled by gravitational self-collapse of gas and dust under ordinary Newtonian/ GR gravity, with no need for an additional "trapping" mechanism; H4 needs to say what observation it predicts that the standard core-accretion picture does not. (ii) The claim that *every* body in space – every planet, asteroid, and ordinary star – has a black hole at its center is directly checked against existing evidence and, as stated, is in tension with it: the Sun's interior structure is measured via helioseismology, Earth's via seismology and its geodetic and magnetic-field behavior (which requires an ordinary convecting liquid-metal core), and none show the signatures a central black hole would produce (anomalous mass-radius relation, an accretion luminosity or steady mass growth, or the tidal/precision-tracking signatures a compact central mass would leave). This is the most directly falsifiable claim among all four hypotheses, and currently the one with the most existing evidence against it as stated; narrowing it – e.g., to large bodies only, or to the very early universe before ordinary structure formed, closer to how the actual PBH hypothesis is framed – would bring it back into contact with open questions rather than settled observations.

4.6 Hypothesis 5: Supermassive Black Holes as Vortices from Colliding, Oppositely-Expanding Regions of Space

Statement (rough form). Ordinary (stellar-scale) black holes are minor turbulence, tears, or inconsistencies in space. Supermassive black holes are categorically different: they form where large regions of space, expanding in opposing directions, collide and create a large vortex. Supermassive black holes are not, in general, of stellar-collapse origin – though that origin is not impossible in principle – and share more in common with cyclones and hurricanes than with collapsed stars.

Formal statement. [PLACEHOLDER: "regions of space expanding in opposing directions" and "collision" need to be defined relative to the standard cosmological picture, in which the expansion rate (the Hubble parameter) is a single, roughly uniform, non-directional quantity at a given cosmic time – not a vector flow field with distinct, opposing currents]

Motivation. Correctly inherits the real seed problem from H3: a single star's collapse cannot directly produce a 10^6–$10^{10}\,M_\odot$ object, so some categorically different formation channel for the largest black holes is a reasonable thing to look for. The vortex picture also has a genuine, if indirect, point of contact with real physics: real (rotating, Kerr) black holes drag spacetime around them (frame dragging / the Lense–Thirring effect, now measured via accretion-disk precession and X-ray spectroscopy), and this rotational behavior is precisely what motivates *analogue-gravity* experiments, in which a real draining-fluid vortex is built in the lab because the mathematics of waves near its drain matches the mathematics of waves near a rotating black hole's horizon (Unruh, 1981; demonstrated with real draining-bathtub vortices showing rotational superradiance). That match is at the level of the wave equations, however, not a claim that spacetime itself is a literal fluid.

What would falsify it / open problems. (i) The standard cosmological model (the homogeneous, isotropic FLRW spacetime, supported by the measured isotropy of the cosmic microwave background and of large-scale structure) describes expansion as a single scale factor increasing over time, not as a flow field with distinct regions expanding in different directions that could collide; H5 needs to state what observation would reveal such oppositely-expanding regions, since none has been reported. *This specifically rules out a large-scale, coherently directional version of colliding expansion* – the CMB-isotropy bound behind it constrains a single, global flow pattern (§2.1), not scattered, local patchiness. A version of H5 built instead on small, randomly-distributed regions with locally differing expansion histories is not excluded by this bound; it instead has to clear the two items immediately below (stress-energy content,

and GR's smoothness assumption) and show it is doing something the already-measured 10^{-5} density perturbations do not already cover. (ii) Even granting a collision, converting the kinetic energy of "colliding space" into spacetime curvature strong enough to form a horizon requires an effective stress-energy content for the colliding regions — the same open requirement flagged for H1 and H4, since GR curves in response to stress-energy, not motion of space considered as already-empty geometry. (iii) The nearest existing, if still speculative, research precedent for "collision forms a black hole" is bubble-collision black-hole formation during an early-universe phase transition, studied in specific inflationary/phase-transition models – but that mechanism is tied to the very early universe and a specific particle-physics scenario, not to expansion generally or to an ongoing, occasional process. (iv) Rotation (frame dragging) in real supermassive black holes is well explained as inherited angular momentum from the matter and smaller black holes that merged and accreted to form them – the standard growth picture H5 sets out to replace – so H5 would need to show what a vortex-of-colliding-space origin predicts that spin-from-accretion-and-mergers does not. (v) *The Bullet Cluster test, run against this specific mechanism.* If colliding space behaved like an ordinary, dissipative fluid – shocking and slowing on contact, the way the Bullet Cluster's hot gas does – a vortex sourced by it should track the gas, not the galaxies; the Bullet Cluster shows the lensing mass tracking the (collisionless) galaxies instead, which would rule this reading out directly. H5 survives this test only by committing explicitly to colliding space passing through itself *nearly without friction* – the same near-collisionless behavior already assumed of, and separately constrained for, particle dark matter. This is a real, available option, not automatically excluded, but it is a specific commitment H5 needs to state, not an automatic property of "space colliding" – a real medium usually has some self-interaction, and asserting otherwise only to match this one observation is circular unless an independent reason for near-frictionlessness is given. (vi) *Mass-energy content, and where it would have to come from.* Point (ii) above already notes that curving spacetime requires a stress-energy source, not just the relative motion of already-empty geometry; stated plainly, this means H5 currently has no quantified energy content for "colliding space" to supply. This is exactly the gap **H7** (§4.8) is attempting to fill from the other direction – if H7's claim that expansion/compression of space carries real, quantifiable energy were established, it would hand H5 precisely the missing ingredient. As things stand, neither hypothesis has independently delivered this number: H5 needs energy content from moving space that H7 has not yet quantified, and H7's own magnitude check (§4.8 discussion) found the effect negligible at the present cosmic epoch for exactly this reason. H5 and H7 should be read as leaning on the same unresolved piece, not as two separately-supported hypotheses.

4.7 Hypothesis 6: Galaxies as Matter Settled into a Vortex's Depression, Replacing Dark Matter

Statement (rough form). A galaxy is matter tracing out the depression in space caused by a central vortex (a supermassive black hole, per H5). Matter settles into this depression. Stars orbit the central vortex at speeds set by the strength and reach of the depression it causes; faster-than-expected orbital speeds at a galaxy's edge come from the vortex's own strength and nature, and dark matter is not required to explain them. Different galaxy morphologies reflect different forms or strengths of the underlying spacetime turbulence.

Formal statement. [PLACEHOLDER: needs an explicit $v(r)$ (orbital speed as a function of radius) predicted from "vortex strength," to compare against the measured rotation curves this hypothesis is meant to explain]

Motivation. Correctly identifies the real observation: galaxy rotation curves are measured to stay flat (roughly constant orbital speed) out to tens of kiloparsecs, far beyond where the visible (stellar and gas) mass alone, via ordinary $1/r^2$ gravity, predicts a declining (Keplerian) curve – this is the original dark-matter evidence (Rubin & Ford, 1970s) and a real target for any competing explanation.

What would falsify it / open problems. (i) *Scale mismatch.* A supermassive black hole's own gravitational reach (its "sphere of influence") extends only a few parsecs to a few tens of parsecs from the galactic center – it dominates stellar orbits only there. Flat rotation curves are measured at tens of *kilo*parsecs, roughly three to four orders of magnitude farther out, where the central black hole's own gravitational pull, by standard $1/r^2$ falloff, is already negligible. For H6 to work, "the vortex" has to mean something with far greater mass and reach than the black hole itself – at which point it is doing the explanatory work dark matter currently does, under a new name, and needs its own mass profile stated and checked against the data the way a dark-matter halo profile is. (ii) *Independent evidence beyond rotation curves.* Dark matter is inferred through several lines of evidence besides rotation curves, each

of which a replacement needs to address: the Bullet Cluster, where gravitational lensing maps show the dominant mass concentration spatially separated from the hot, X-ray-emitting gas that holds most of the ordinary matter in the colliding clusters – direct evidence for a gravitating component not tied to where the normal matter sits; the relative heights of the cosmic microwave background's acoustic peaks, which fix the ratio of ordinary to total matter in the very early universe, long before galaxies or their central black holes existed; and the growth rate of large-scale structure, which needs the extra gravitational pull of a non-baryonic component to form galaxies within the age of the universe at all. A vortex tied to a galaxy's own central black hole has no obvious bearing on any of these three, since two of them predate galaxy formation and one (Bullet Cluster) is at cluster, not single-galaxy, scale. (iii) H6's closest existing competitor is Modified Newtonian Dynamics (MOND; Milgrom, 1983), which changes the force law itself at low accelerations and reproduces many individual galaxies' rotation curves strikingly well without extra matter – but is also known to struggle with exactly the cluster-scale and CMB evidence in (ii). If H6 is a vortex-flavored version of the same idea, it inherits MOND's real successes but should also expect, and address, MOND's known difficulties rather than treating rotation curves as the only test. (iv) *Galaxy-to-galaxy diversity.* Galaxies of similar visible (and presumably similar central-black-hole) mass are observed to need very different amounts of inferred dark matter – some ultra-diffuse galaxies need a large extra-mass fraction, while a few reported galaxies appear to need very little. A mechanism tied tightly to a single central vortex's mass needs to explain this spread; the standard dark-matter picture explains it as galaxies having assembled with different amounts of dark-matter halo relative to their visible mass, through different formation histories.

4.8 Hypothesis 7: Energy as a Function of the Expansion or Compression of Space

Statement (rough form). When space expands, it loses energy in some form; when it compresses, energy becomes available. Particle creation is one carrier of this exchanged energy. (An earlier statement of this hypothesis went on to claim that this energy "should be capturable"; as the falsify discussion below shows, the established mechanisms this draws on are conversions, costing at least as much energy as they release, not net sources, so that extraction claim is dropped and is not part of H7 as stated here.)

Formal statement.

$$\frac{dE}{d\lambda} \stackrel{?}{=} f(\theta, \dot{\theta}), \qquad f \text{ unspecified} \tag{14}$$

Motivation. This is the best-grounded of the new hypotheses in this paper, because the general phenomenon it describes is real and already established in three separate, serious contexts: cosmological particle creation (Parker, 1968), in which the expansion of spacetime stretches quantum vacuum fluctuations into real particles – the accepted mechanism by which inflation seeded the density variations later seen in the cosmic microwave background; reheating at the end of inflation, where the energy driving that expansion converts directly into the real particles that went on to form everything in the universe; the dynamical Casimir effect, in which rapidly changing the boundary of a vacuum region has been experimentally shown to create real photons (Wilson et al., 2011); and Hawking radiation, in which the changing, compressing geometry around a collapsing mass converts vacuum fluctuations into real, radiated particles (Hawking, 1974). H7 is, in effect, the claim that these four known mechanisms are instances of one general principle, tying particle/energy creation to θ (or its rate of change) rather than requiring each to be re-derived separately.

What would falsify it / open problems. (i) *Magnitude, for the present epoch.* Cosmological particle creation becomes significant only when the expansion rate is comparable to the particle's own natural (Compton) frequency, $mc^2/\hbar$. Today's Hubble rate is approximately $2 \times 10^{-18}\,\mathrm{s}^{-1}$; even the lightest known particle, a neutrino of mass $\sim 0.1\,\mathrm{eV}$, has a Compton frequency of order $10^{14}\,\mathrm{s}^{-1}$ – a gap of roughly 32 orders of magnitude. The one episode when this gap closed was inflation, not the present day, so H7's effect, however real in principle, is not occurring at any detectable rate in the universe as it expands right now. (ii) *Net extraction and energy conservation.* In every established instance – the dynamical Casimir effect, Hawking radiation, inflationary reheating – the vacuum acts as a conversion mechanism, not a net energy source: driving the dynamical Casimir effect costs energy to move the boundary, at least as much as is recovered as photons; Hawking radiation draws down the radiating black hole's own mass as its price; reheating converts the energy of a specific, already-present field (the inflaton), it does not create energy from nothing. "This energy should be capturable" is, in its strongest reading, a claim that energy conservation can be circumvented, which is a substantially harder barrier to

clear than any other objection raised in this paper – $\dot{G}/G$ bounds, CMB isotropy, or interior-structure data can all, in principle, be satisfied by a sufficiently small or well-hidden effect; conservation of energy is not a bound to be satisfied by smallness, but a constraint on the bookkeeping itself. H7 is best read, for now, as the unification claim in its *motivation* (one principle behind four known phenomena) without the extraction claim, unless and until a specific mechanism is proposed that identifies exactly what pays the energy's cost, the way the black hole's own mass pays for Hawking radiation.

4.9 Hypothesis 8: Black Hole Formation as a Core Sinking Into Time

Statement (rough form). When a collapsing star's core becomes too massive for anything to support it against its own gravity, it does not collapse to a literal point instantly; it becomes a dense, finite-sized core that sinks into space gradually, the way a ball bearing sinks slowly through honey. As it sinks, it becomes steadily more a feature of the time dimension than of space, generating a vortex as it goes. The point where the core finally disappears, having sunk completely, is the singularity – not a pre-existing infinitesimal point the core falls into, but the gentle closing of the gap behind it.

Formal statement. [PLACEHOLDER: needs the core's radius as an explicit function of proper time during collapse, $R_{\rm core}(\tau)$, and a statement of what, physically, sets its minimum (non-zero) size]

A formal link to H1's elasticity: the threshold at which space's strength is overwhelmed. H8's core intuition – that space can support a body only up to some limit, and lets go beyond it – has an exact, already-established counterpart: Buchdahl's theorem (1959) proves that no static, pressure-supported configuration of mass M can exist at all once its radius drops below

$$R_{\rm crit} \;=\; \frac{9}{8}\cdot\frac{2GM}{c^2} \;=\; \frac{9GM}{4c^2} \;=\; \frac{9Mc^2}{32\pi\Sigma}, \tag{15}$$

using §3.2's $\Sigma = c^4/8\pi G$ in the last step. Written this way, Σ – H1's elasticity – sits directly in the threshold: a stiffer space (larger Σ) only fails at a smaller, more compact $R_{\rm crit}$, while a softer space (smaller Σ) gives way at a larger, less-compact radius. This is a precise, checkable version of "space's strength to bear a body": below $R_{\rm crit}$, no amount of pressure can hold the core up, regardless of its composition, and collapse – H8's sinking – becomes unavoidable.

What this does and does not resolve. This closes exactly half of open problem (ii) below: it gives a specific, Σ-linked *onset* condition – the point at which space's static-support capacity is overwhelmed – rather than leaving "too massive for anything to support it" as a qualitative statement alone. It does *not* say what finite size $R_{\rm core}$ the sinking core settles at afterward; that remains the genuinely open question this subsection already points to (Planck stars, fuzzballs), and Buchdahl's bound is silent on it by construction – it marks where support fails, not where collapse stops.

Motivation. More of this maps onto real, already-established or already-seriously-studied general relativity than any other hypothesis in this paper:

- *"Too heavy for anything to support it"* is exactly the real physics of collapse: electron degeneracy pressure supports a white dwarf only up to the Chandrasekhar limit ($\sim 1.4\,M_\odot$); neutron degeneracy pressure supports a neutron star only up to the Tolman–Oppenheimer–Volkoff limit (~ 2–$3\,M_\odot$); beyond either limit, no known form of pressure halts the collapse.
- *"Sinking into space, like a ball bearing in honey"* is a good intuitive restatement of the *river model of gravity* already introduced for H2 (Painlevé, 1921; Gullstrand, 1922): in that already-standard, already-equivalent- to-Schwarzschild picture, space itself is pictured as flowing inward toward the mass, reaching the speed of light exactly at the horizon – matter does not so much fall *through* space as get carried inward *by* it, very much like a bearing settling through a viscous fluid.
- *"Becomes more a feature of the time dimension"* is not a loose metaphor – it is a precise, well-known, and somewhat startling feature of the Schwarzschild solution itself: inside the horizon, the roles of the radial and time coordinates swap, so that moving toward the center becomes as unavoidable, and as direction-less, as moving forward in time is outside it. "Falling toward the center is like being swept forward in time" is a standard way this fact is described in general relativity courses, not a new claim.
- *"A dense core, not an infinitely small point"* tracks a genuinely open and actively studied question: whether a true, infinite-density singularity actually forms (as classical GR's singularity theorems,

Penrose and Hawking, predict under very general conditions) or whether quantum-gravitational effects halt collapse at some very small but finite size, replacing the singularity with something else – proposals include "Planck stars" (Rovelli & Vidotto, 2014) and "fuzzballs" (Mathur, 2005, in string theory). H8's finite core is a plain-language version of this exact open question, not a departure from it.

What would falsify it / open problems. (i) *The vortex needs its own, separate ingredient.* The river model's inflow for a non-rotating collapse is purely radial – no swirling at all. A genuine vortex (frame dragging, the Kerr metric's rotation) requires the collapsing core to carry angular momentum, inherited from the progenitor star's own spin; it is not an automatic consequence of "sinking" by itself. H8 needs to state explicitly that the vortex component depends on, and scales with, the progenitor's rotation – without that, a non-rotating collapse in this picture should produce no vortex at all, which is worth checking against whichever of H3/H5's claims this is meant to connect to. (ii) *The finite-core size is not yet a specific claim – onset now covered, the settled minimum still is not.* The paragraph above gives a specific, Σ-linked answer to *when* collapse becomes unavoidable ($R_{\rm crit} = 9Mc^2/32\pi\Sigma$), but not to *what floor* $R_{\rm core}$ settles at once collapse proceeds past that point. Planck stars and fuzzballs each make a specific proposal for that remaining question, at what scale (roughly the Planck length, or a horizon-scale structure, respectively); H8 still needs its own specific answer to it before it can be checked against, or distinguished from, those two existing proposals.

A bounded extension: time's direction tipping, not just slowing, inside the vortex. H8 already uses one precise, established fact about the Schwarzschild interior – that the radial and time coordinates swap roles, so infall becomes as unavoidable as moving forward in time. Real *rotating* spacetimes go further than this in a way worth adding explicitly, rather than leaving as background intuition: inside a Kerr black hole's ergosphere, and throughout Gödel's rotating-universe solution (1949, already cited for H2b), light cones genuinely *tip* – the locally-available "forward in time" direction reorients relative to a distant observer's, to the point that Gödel's solution permits closed timelike curves outright. This is real, if exotic, general relativity, not speculation reaching beyond it. Proposed addition, scoped narrowly: wherever H8's sinking core carries genuine angular momentum (the still-open ingredient from point (i) above), the resulting vortex should exhibit this same light-cone tipping, not merely ordinary time dilation – giving H8's "more a feature of time than of space" language a sharper, rotation- specific reading it does not yet have. This is explicitly *not* a claim that time tips everywhere, or in any region without real rotation; it is bounded to the same vortex regions H5 and H8 already propose, and inherits H8's own open item (i) as a precondition: no stated angular momentum means no tipping, exactly as it currently means no vortex.

4.10 Hypothesis 9: Gravity's Weakness as Leakage into a Hidden Dimension

Statement (rough form). Gravity is not intrinsically different in kind from the other three forces – it is diluted. Space has at least one dimension beyond the four (three space, one time) this paper has otherwise assumed throughout, and gravity alone is free to spread into it, while matter and the other forces remain confined to our own four. What we measure as gravity's strength – including the strain of a passing gravitational wave – is a diluted remnant of its true strength in that larger setting, not gravity's actual, undiminished magnitude.

Formal statement.

$$\Sigma_0 \overset{?}{\sim} \Sigma_{\rm bulk} \cdot \left(\frac{L}{\ell_*}\right)^n , \tag{16}$$

where $\Sigma_{\rm bulk}$ is gravity's coupling in the full $(4+n)$-dimensional bulk, L is the extra dimension(s)' size, ℓ_* a fundamental length scale, and n the number of extra dimensions – schematic, modeled directly on the ADD relation $M_{\rm Pl}^2 \sim M_*^{n+2} L^n$, not yet derived within this paper's own formalism. The measured Σ_0 (§3.2) is large – gravity weak – because $L \gg \ell_*$, not because gravity's bulk coupling differs in kind from the other forces'.

Motivation. The hierarchy problem is real and unsolved: gravity is measured to be roughly 10^{36} times weaker than electromagnetism between two protons, with no accepted explanation for why gravity specifically is so feeble. H9 adopts an existing, actively studied answer to this (§2.3) rather than proposing a new one, and connects it to this paper's own language: Σ_0's measured, constant value (§4.1, point 1)

is reframed as a 4D-effective, diluted projection of a deeper bulk constant, not a contradiction of H1's result but an additional layer underneath it.

What would falsify it / open problems. (i) n and ℓ_* are not yet specified; until they are, H9 makes no numerical prediction. (ii) It is already tightly bounded regardless: sub-millimeter torsion-balance tests of Newton's law already exclude any such dimension larger than roughly 40–50 μm, and collider searches for missing-energy gravitons constrain it further – so the more dramatic reading this hypothesis was originally reaching for (gravitational waves "huge" at any accessible scale) is specifically excluded; only a very small, already-tightly-bounded version remains available. (iii) H9 must still recover Σ_0's measured constancy in every regime already checked (§4.1, point 1; §5) – dilution must be uniform enough, or L stable enough, that no detectable variation in Σ_0 results. (iv) H9 is, at present, an adopted explanation, not a new one; its distinct content, if any, would need to show what this paper's own formalism (Σ, $\mathcal{I}$) adds to the existing ADD/Randall–Sundrum picture rather than merely restating it.

4.11 Hypothesis 10: Spacetime's Elasticity as an Induced, Not Fundamental, Property

Statement (rough form). H1's elasticity constant Σ_0 – the stiffness that resists curvature – is not a bare, fundamental constant of nature, assumed and then measured, but an *induced* quantity: the aggregate, macroscopic residue of the quantum vacuum fluctuations of every matter field, summed up to whatever energy scale new physics takes over. Nothing is added to spacetime's own structure to produce this stiffness; it emerges the way a liquid's surface tension emerges from the sum of countless molecular cohesive forces, not from any dedicated "surface tension field." An ant standing on water does not dimple the water because water has a fundamental stiffness built into it at the start; the dimple's restoring force is the aggregate of billions of hydrogen bonds resisting the local disturbance. H10 proposes spacetime's resistance to curvature works the same way, with the quantum vacuum playing water's role and mass playing the ant's.

Formal statement. Following Sakharov (1967), integrating out the vacuum fluctuations of N quantum field species up to an ultraviolet cutoff Λ generates, among other terms, an effective Einstein–Hilbert term in the gravitational action with coefficient

$$\Sigma_{\text{induced}} \sim \frac{N}{8\pi}\frac{\Lambda^2}{\hbar c}, \tag{17}$$

with the same units (force) as $\Sigma_0 = c^4/8\pi G$ (§3.2); matching Σ_{induced} to the measured Σ_0 for some stated N and Λ is the concrete, numeric content of this hypothesis, carried out below. N and the prefactor are schematic – the exact coefficient depends on the spin and multiplicity of the fields summed and on the regularization scheme used, neither of which this paper derives from first principles; it is quoted here as an order-of-magnitude relation, in the same spirit as H9's own schematic ADD relation.

Motivation. Three independent precedents support treating a macroscopic "stiffness" as emergent from microscopic structure rather than fundamental: (a) the Casimir effect (already cited under P1, §2), a measured, real force arising directly from vacuum fluctuations between two conducting plates – direct laboratory evidence that the vacuum is not simply empty, and that its fluctuations exert real, calculable mechanical effects; (b) ordinary surface tension, which is not a fundamental force but the macroscopic sum of molecular cohesion – water's $\gamma \approx 0.072\,\text{N/m}$ at room temperature is derivable, in principle, from its molecules' hydrogen-bond strength, and different liquids (mercury, $\gamma \approx 0.49\,\text{N/m}$; ethanol, $\gamma \approx 0.022\,\text{N/m}$) have different tensions because they are made of different things, not because "surface tension" is itself an irreducible ingredient of nature; (c) condensed-matter elasticity generally (already invoked for H1b, §4.2), where macroscopic elastic moduli are routinely derived from microscopic lattice structure rather than postulated. H10 asks whether Σ_0 is the gravitational instance of the same pattern: not an assumption this paper (or GR) is entitled to make for free, but a number that should, in principle, be computable from the actual roster of fields nature contains. (Note the scope, since the ship/water picture motivating H2c and the ant/water picture motivating H10 are not the same claim: H2c's ship is about a body moving *through* an already-curved medium; H10's ant is about why the medium has any stiffness to curve against in the first place.)

What would falsify it / open problems. (i) *The central, decisive problem: cutoff sensitivity.* Using the Standard Model's own field content ($N \sim 100$, order of magnitude) and its own natural energy

scale, the electroweak scale ($\Lambda \sim 1\,\mathrm{TeV} = 1.602 \times 10^{-7}\,\mathrm{J}$), gives

$$\Sigma_{\mathrm{induced}} \sim \frac{100}{8\pi} \times \frac{(1.602 \times 10^{-7}\,\mathrm{J})^2}{(1.055 \times 10^{-34}\,\mathrm{J\,s})(3 \times 10^{8}\,\mathrm{m/s})} \approx 3 \times 10^{12}\,\mathrm{N},$$

roughly *thirty orders of magnitude* short of the measured $\Sigma_0 \approx 4.98 \times 10^{42}\,\mathrm{N}$ ($\Sigma_0/\Sigma_{\mathrm{induced}} \approx 1.5 \times 10^{30}$). Equivalently, induced gravity built from Standard-Model physics alone predicts gravity should be roughly 10^{30} times *stronger* than it actually is. The only cutoff that reproduces the correct Σ_0 is $\Lambda \sim$ the Planck scale itself – but the Planck scale is *defined from* G, so using it as an input is circular, not a derivation. This is not a minor residual; as stated, H10 does not yet work. (ii) *This is very likely the same problem as H9's.* H9's hierarchy problem (10^{36}: gravity vs. electromagnetism between two protons) and H10's cutoff problem (here, $\sim 10^{30}$) are, in all likelihood, two restatements of one underlying puzzle – why the scale at which gravity becomes strong ($\sim M_{\mathrm{Pl}}$) sits so far above the scale at which the other forces live ($\sim 1\,\mathrm{TeV}$) – rather than two separate open items this paper can claim independent progress on. Resolving one without the other would be a strong signal that the resolution is not yet the right one. (iii) *Unlike H9, H10 does not import an existing, independently-developed fix.* H9 at least adopts an established answer to the hierarchy problem (ADD/Randall–Sundrum); H10 imports a different existing problem (Sakharov's own, known since 1967) without yet importing any proposed fix to it (low-scale supersymmetry's partial cancellations, or other beyond-Standard-Model content that could, in principle, bring N or Λ into line) – doing so honestly is the next item of work, not a detail to wave past. (iv) H10 inherits H1's own open item unchanged: it explains why spacetime has *some* stiffness (because something must resist being disturbed, and ordinary vacuum fluctuations are a real, measured phenomenon that does this for other systems), but even a fully successful version would still not explain why matter couples to curvature via the specific minimal-coupling structure GR assumes in the first place – that structural assumption is inherited from ordinary QFT-in-curved-spacetime unchanged, not derived by this hypothesis.

Conclusion a reader can draw: H10 is the right *kind* of answer to "why does spacetime have elasticity at all" – it has real, independently-established precedent (the Casimir effect, surface tension, condensed-matter elasticity) for treating a macroscopic stiffness as an emergent residue of microscopic structure rather than a given. But as a specific numeric claim about *this* universe's Σ_0, it currently fails its own check by roughly thirty orders of magnitude, for essentially the same structural reason H9's hierarchy problem remains open. H10 is kept here as a motivated, honestly-failing hypothesis, not a working mechanism – it names the right question and the right general shape of answer, without yet supplying the missing physics (new fields, a different cutoff, or some other resolution of the hierarchy problem) that would close the gap.

4.12 Taxonomy of Spacetime Inconsistencies

H3 through H6 each reach for the same underlying idea – that spacetime is not smooth, and that departures from smoothness come in different forms – under different names (a tear, a depression, a vortex, a crack or channel). Treating these as named *species* of one genus, "spacetime inconsistency," is a workable move, but only if each species is given its own criteria up front. Otherwise the genus absorbs any new observation after the fact without having predicted it beforehand, which trades away falsifiability rather than gaining explanatory power. The table below is a first attempt at those criteria; every row currently has open entries that need a decision before the taxonomy does real work.

Species	Used by	Horizon?	Mass-energy source?	Scale	What would test it
Sphere (classical BH)	H3, H4, H5, H6 (galactic core)	Yes	Yes – standard GR collapse or accretion	Stellar to $10^{10}\,M_\odot$	Already tested: matches GR/Kerr predictions, event-horizon imaging (EHT)
Vortex	H5 (SMBH origin), H6 (galaxy-scale reach)	[PLACEHOLDER: decide]	[PLACEHOLDER: what sources the rotation?]	Galactic (∼kpc–Mpc)	Must reproduce flat rotation curves *and* Bullet Cluster, CMB peaks (§4.7)
Depression (micro)	H4	No (by H4's own revision, §4.5 discussion)	[PLACEHOLDER: ordinary local mass, or spontaneous?]	Sub-planetary to planetary	Must not exceed Hawking-evaporation or runaway-accretion bounds for its assumed mass (§4.5)
Crack / channel	H6 (cosmic web framing)	[PLACEHOLDER: decide]	[PLACEHOLDER: decide]	∼10–100 Mpc (filaments)	Must reproduce filament/void statistics already matched by gravity + cold dark matter (Zel'dovich approximation; Millennium/IllustrisTNG simulations)

The two columns most worth settling first are *Horizon?* and *Mass-energy source?*, since they are what let H3–H6 borrow (or not) from established black-hole physics without equivocating between species: a species with a horizon inherits Hawking radiation and accretion-growth constraints and must be checked against them, as the Sphere row already is; a species without one cannot claim to be "technically a black hole" when that label is convenient elsewhere in the paper. The *Mass-energy source* column is the same open question raised for H1 (§4.1) and H2 (§4.3): under standard GR, curvature is sourced by stress-energy, so any species claimed to arise without a stated mass-energy source is proposing an amendment to the field equations themselves, not a new kind of black hole, and should be argued for at that level.

5 Consistency Checks

5.1 Recovery of known limits

H1's scoped formal statement (§4.1) reduces to Newtonian gravity, $g = GM/r^2$, by construction: Σ_0 plays G's role exactly, with no new free parameter at this stage. That recovery is not merely formal – it has been checked numerically against five independent bodies.

Five-body check: Earth, Mars, Sun, Jupiter, Saturn. Define $A := g/(H_0^2 R)$ for each body, using its measured surface gravity g, radius R, and today's Hubble parameter $H_0 \approx 2.18 \times 10^{-18}\,\mathrm{s}^{-1}$. If elasticity contributed anything beyond ordinary Newtonian gravity, A would need an independent, per-body explanation. Instead:

Body	ρ (kg/m^3)	g (m/s^2)	$A = g/(H_0^2 R)$
Earth	5514	9.81	3.24×10^{29}
Mars	3933	3.71	2.30×10^{29}
Sun	1408	274	8.29×10^{28}
Jupiter	1326	24.79	7.46×10^{28}
Saturn	687	10.44	3.77×10^{28}

A is not constant – it spans a factor of ∼8.6 across these five bodies – but it is not independent new physics either: it tracks each body's density *exactly*, since algebraically $A = (4\pi G/3H_0^2)\rho$, with H_0^2 cancelling completely once A is multiplied back through to recover g. The ratio $A/A_\oplus$ matches $\rho/\rho_\oplus$ to within a few percent for every body in this table (the residual is measurement rounding and non-sphericity for the gas giants, not a missing term). **Conclusion:** H_0 plays no role in local gravity at all, and every number here is exactly what $g = GM/r^2$ already predicts – the recovery of the Newtonian limit is confirmed, with zero residual left for elasticity to explain beyond G's own, already-measured value.

Shell-theorem check. H1's claim that bend depends on mass and distance, not density and radius independently (§4.1, point 2), was checked directly: Earth's actual mass and radius give $g = 2.454\,\mathrm{m/s^2}$ at $r = 1.274 \times 10^7$ m; a hypothetical body of the same mass but radius equal to that same r (density $\approx 689\,\mathrm{kg/m^3}$) gives the identical $g = 2.454\,\mathrm{m/s^2}$ at its own surface. Density and radius carry no information beyond what M and r already fix.

Birkhoff's theorem as a bound on Σ and on fluidity. Birkhoff's theorem (1923) shows that spherical symmetry plus the *vacuum* field equations with a fixed coupling Σ_0 force the exterior field of any non-rotating mass to be static and exactly Schwarzschild – a result that makes no reference to, and needs no input from, the kinematic triplet $(\theta, \sigma_{\mu\nu}, \omega_{\mu\nu})$ of §3.2: those describe how observers move through a spacetime once it is fixed, not an ingredient in fixing it. The theorem's premises, however, assume exactly what H1 and H2 propose to relax: a constant Σ, and no curvature contribution from expansion or spin beyond what mass-energy already supplies through the Einstein equations. If $\Sigma(x)$ varies, or if θ or $\omega_{\mu\nu}$ contribute to curvature independently of local mass-energy, Birkhoff's conclusion is no longer guaranteed: the field around an isolated, non-rotating mass need not stay static or purely Schwarzschild in form. This is a real, independent consistency requirement, not a restatement of the $\dot{G}/G$ and cosmic-rotation bounds already noted in §4.1 and §4.3: the solar system's gravitational field has itself been measured to extraordinary precision (planetary ranging, light bending, perihelion precession) and matches the static Schwarzschild/Kerr prediction with no detected extra time-dependence or unexpected multipole structure. **Any variation proposed for Σ, and any curvature sourced by θ or $\omega_{\mu\nu}$ beyond the mass-energy term in the Raychaudhuri equation, must be small enough in the solar-system regime to leave Birkhoff's static, Schwarzschild-shaped conclusion intact there** – in addition to, not instead of, matching the measured value and constancy of Σ_0 itself.

5.2 Dimensional analysis and units

Σ. $\Sigma_0 = c^4/8\pi G$: $[c^4] = \mathrm{m^4/s^4}$, $[G] = \mathrm{m^3/(kg\,s^2)}$, so $[\Sigma_0] = (\mathrm{m^4/s^4}) \cdot (\mathrm{kg\,s^2/m^3}) = \mathrm{kg\,m/s^2} = \mathrm{N}$ – a force, exactly matching the already-standard "stiffness of spacetime" reading of this constant (§3.2) and consistent throughout H1/H1b.

$g = GM/r^2$. $[GM] = \mathrm{m^3/s^2}$, divided by $[r^2] = \mathrm{m^2}$ gives $\mathrm{m/s^2}$, an acceleration – checks out in every numeric table in this section.

$\mathcal{I}(x)$ – an item this check surfaced and that has since been corrected at the definition. θ, $\sigma_{\mu\nu}$, $\omega_{\mu\nu}$ each carry units of inverse time (an expansion/shear/rotation *rate*), so $(\theta - \bar{\theta})^2$ has units $\mathrm{s^{-2}}$. For $\mathcal{I} := \ell^2\sqrt{(\theta-\bar{\theta})^2 + \cdots}$ (§3.2) to come out dimensionless as claimed, ℓ^2 must carry units of $\mathrm{s^2}$ – i.e. ℓ is a reference *time*, not a length, despite the symbol – and §3.2's definition now states this explicitly rather than calling ℓ a length scale. §3's broader units convention is still an open placeholder, so ℓ's specific numerical value (what time scale, fixed by what physical consideration) remains open and should be settled before $\mathcal{I}$ is used quantitatively in H1b or the taxonomy (§4.12) – but the dimensional inconsistency itself, between calling ℓ a length at its definition and needing it to be a time here, is resolved.

5.3 Known observational constraints

Passed (positive recoveries, not just non-contradictions). H1's scoped formal statement matches two independent, high-precision measurements exactly: the GPS gravitational time-dilation correction ($\sim 45.7\,\mu$s/day, from $\Delta(GM_\oplus/rc^2)$ between ground and satellite altitude) and the solar gravitational redshift ($\approx 2.12 \times 10^{-6}$, from $GM_\odot/R_\odot c^2$). Neither required any adjustment to reproduce.

Constraints Σ (constant) already satisfies. Lunar-laser-ranging and pulsar-timing bounds on $\dot{G}/G$ (parts in 10^{11}–10^{12} per year) directly bound H1's constant-Σ claim (§4.1, point 1) and are satisfied by construction, since H1 proposes no variation. The Cassini Shapiro-delay bound on the Brans–Dicke parameter ($\omega > 40{,}000$; Bertotti, Iess & Tortora, 2003) is the relevant constraint on H1b specifically (§4.2): any $\lambda\, h(\mathcal{I})$ term must stay within whatever this bound translates to in the solar-system regime, where $\mathcal{I}$ is small.

A constraint an earlier version of this framework failed, and was revised in response. An amplification factor tying local gravity to H_0 directly was tested against Big Bang nucleosynthesis: the Hubble parameter at BBN ($\sim 0.5\,\mathrm{s}^{-1}$) exceeds H_0 by a factor of $\sim 2.3 \times 10^{17}$, so a gravity term scaling as H^2 would have been $\sim 5 \times 10^{34}$ times stronger at BBN – grossly inconsistent with the observed light-element abundances, which match standard G to within ∼10–20%. This is why H_0 does not appear in H1's final formal statement: it cancels out of the recovered formula algebraically in any case (§5.1), and taking it as a literal, real-time ingredient is independently ruled out by this bound. Recorded here as a constraint the hypothesis's earlier form failed and was corrected against, not one it currently needs to satisfy.

Constraints not yet checked. Equivalence-principle (Eötvös-type) tests and gravitational-wave speed/waveform constraints bear on H2's and H1b's claims specifically and have not yet been worked through numerically; flagged here as the next items for this subsection. Light bending and Mercury's perihelion precession, which belonged on this list through earlier revisions, have since been evaluated numerically and removed from it: H2c's deflection split, evaluated for the Sun, reproduces the measured $1.75''$ solar deflection, and the same mechanism's perihelion-advance formula reproduces Mercury's measured $42.98''$/century excess precession (both in §4.3's numeric checks, immediately following H2c's formal statement) – the standard GR formulas transcribed and evaluated, not independent derivations from Σ.

5.4 Internal consistency

Density is not an independent degree of freedom alongside elasticity. The five-body table in §5.1 shows $A = (4\pi G/3H_0^2)\rho$ exactly – the quantity that seemed to vary body-to-body is density itself, relabeled, not an independent confirmation that "elasticity" is doing new explanatory work. This is logged here as a resolved internal-consistency question for H1: constant Σ plus varying ρ fully accounts for every number checked, with nothing left over.

"Bend" is ambiguous between two quantities with different shapes, and H1's language should specify which. Gravitational force/ acceleration falls as $1/r^2$; gravitational time dilation (the potential) falls as $1/r$. These are not the same curve: at a 1% threshold of surface value, the force well closes at $10\,R_\oplus$ while the time well remains open out to $100\,R_\oplus$, and the gap between them grows linearly with r (the ratio is $r/R_\oplus$ itself). H1's informal statement ("the amount of bend") should be read as the force/curvature sense by default, consistent with its formal statement, with the time-dilation sense flagged as a distinct quantity sourced by the same M whenever both are discussed together (as in H2's thought experiment, §4.3).

Absolute well depth and local gradient are different questions, and a hypothesis using "influence" needs to say which it means. Earth's own time-dilation depth is smaller than the Sun's ambient depth at Earth's orbit by a factor of ∼14 – by absolute depth, Earth's influence effectively ends within its own radius. Yet Earth's *gradient* (the quantity GPS clocks and tidal forces actually respond to) dominates the Sun's gradient by a factor of ∼300 at satellite altitude, and remains dominant out to the Hill sphere ($\sim 1.5 \times 10^6$ km, $\sim 235\,R_\oplus$) – the same boundary recovered independently from a tidal-balance argument ($GM_\oplus/r^2 = 2GM_\odot r/a^3$). H2 and H8's "sinking into space/time" language should be read as a gradient claim, not an absolute-depth claim, given this result; stated here so later revisions of H2/H8 inherit the distinction rather than re-deriving it.

Compactness has a sharp internal-consistency boundary, already established, that connects H1 directly to H8. Compressing a fixed mass M into a smaller radius R leaves the field unchanged for any $r > R_{\mathrm{old}}$ (shell theorem) but raises the surface value as $1/R^2$ and the surface time-dilation depth as $1/R$ – a steeper, closer-in bend, with elasticity itself untouched throughout. This

process has a hard stop: once $R \leq R_s = 2GM/c^2$ (for the Sun, $R_s \approx 3$ km; worked Earth-mass example in §4.1 discussion gives $R_s \approx 8.9$ mm), no pressure – not thermal, not electron- or neutron-degeneracy – can balance gravity any further, which is exactly the Chandrasekhar/TOV boundary already catalogued in §2.1. This is the precise, checked link between H1's compaction result and H8's collapse narrative: H1 describes the approach to R_s as a smooth tightening of the bend; H8 describes what happens once R_s is reached. Logged here so H8's eventual revision can cite this derivation rather than re-deriving the threshold.

6 Predictions

This paper's own standard, applied to itself: a prediction needs a regime and a sign, not just a word like "deviation." The minimum set below is what currently meets that bar; several hypotheses (H3, H4, H5, H6 as stated) do not yet have an entry here because they have not yet produced one, which §7 states plainly rather than papering over with a placeholder prediction.

- **H1b.** Any solar-system or binary-pulsar residual from $\lambda\, h(\mathcal{I})$ must sit under the Cassini-derived bound on ω (§4.2) wherever $\mathcal{I}$ is small; the open prediction is a stated, nonzero deviation specifically in a regime that bound does not reach – near a compact, high- curvature core (H3/H4/H8), or in a cosmic-web filament – with a sign and magnitude given in advance of looking.
- **H2-engine.** As a global origin story for Earth's field, this paper's own numbers (§4.3; cH_0 roughly ten orders of magnitude short of $9.81\,\mathrm{m/s^2}$, and $H_0^2 R \approx 3 \times 10^{-29}\,\mathrm{m/s^2}$ – H_0^2 itself is a rate, s^{-2}, not an acceleration, so it is converted via Earth's radius R before comparing to g – roughly thirty orders short) rule the engine out entirely. **H2-revised** (curvature itself as the cause, kept) leaves no prediction to state beyond the geodesic equation's existing, already-standard $-\nabla\Phi$ field – i.e., no prediction beyond GR survives from either version of H2 as currently written.
- **H6.** An explicit $v(r)$ derived from the proposed vortex's strength and reach, plotted against a measured rotation curve at $r \sim 10\,\mathrm{kpc}$, together with an explicit statement of what the same mechanism predicts for the Bullet Cluster's mass/gas offset. Absent both, H6 makes no prediction distinguishable from a renamed dark-matter halo.
- **H7.** A specific relation between particle/energy creation and θ that reproduces the Parker or Hawking rate in its respective established limit, with no net-energy- extraction term – the unification claim stated as a formula, not only as a motivation (§4.8).
- **H8.** $R_{\mathrm{core}}(\tau)$'s minimum set by a stated physical scale (as Planck stars and fuzzballs each already do for their own proposals), and the vortex's strength stated as proportional to the progenitor's own angular momentum, vanishing identically when that spin is zero.
- **H10.** Unlike H3–H6, this entry is not absent for lack of a number – it already has one, and the number fails: using the Standard Model's own field content and energy scale ($N \sim 100$, $\Lambda \sim 1\,\mathrm{TeV}$; §4.11), $\Sigma_{\mathrm{induced}} \approx 3\times10^{12}\,\mathrm{N}$ against a measured $\Sigma_0 \approx 4.98\times10^{42}\,\mathrm{N}$ – short by roughly thirty orders of magnitude. The open prediction this hypothesis still owes is a stated resolution (additional field content, a different cutoff, or a specific beyond-Standard-Model mechanism) that closes this gap without being fit to the answer in advance – i.e., the same bar H9 already clears by citing an existing program, which H10 has not yet done.

7 Discussion

7.1 What this paper does and does not claim

This paper does not claim to have found a new source of gravity, a new source of energy, or a replacement for dark matter. Where a hypothesis's own numeric check came out against it (H2's magnitude and shape problems, §4.3; H6's scale mismatch, §4.7; H4's tension with interior-structure data, §4.5; H10's roughly-thirty-order cutoff-sensitivity problem, §4.11), that result is accepted here as a result, not argued around. What this paper does claim is narrower: that treating space as a real medium with elasticity, fluidity, and irregularity (P1–P3, §2) is a coherent starting point that recovers standard gravity exactly where it must (H1, scoped; §5), correctly identifies several genuinely open questions in current astrophysics

(the SMBH seed problem behind H3/H5; the hierarchy problem behind H9 and, very likely, H10 as well; the vacuum-energy-conversion pattern behind H7), and in at least one case (H8) restates already-serious GR content more plainly than it is usually presented. The value on offer, honestly, is mostly in having specific numbers to check against – several of which came out negative – not in a finished alternative to GR.

Kept, as live and developed claims. **H2a**, in its revised form (**H2-revised** in §6's labeling): gravity as the felt consequence of spacetime curvature itself (§4.3), with the widening gap between neighboring local clock rates as curvature deepens constituting the acceleration ($a = -\partial_r \Phi$, the Newtonian limit of the geodesic equation) – consistent with the GPS correction, Chou et al.'s clock data run in reverse, and Earth's tidal gradient, and adopted as this paper's defining statement of gravity, with the explicit caveat that those three checks are one potential, $\Phi = -GM/r$ for Earth, read three ways, not three independent confirmations, and that the statement itself is a scoped recovery of standard GR rather than content beyond it. **H2b**, kept specifically as the demonstration of what this definition excludes: a centrifugal driver reproduces the same felt sensation but carries zero true, frame-independent curvature, and so does not qualify as gravity under H2a's own definition. **H1**, scoped exactly as §4.1 states it: as vocabulary and a checked recovery, not yet as independent content. **H7**, as a unification proposal aimed at θ, with the net-extraction claim explicitly dropped (§6). **H8**, as the strongest hypothesis in the set, with its two open items (angular momentum, R_{core}'s scale) named rather than assumed away. **H9**, kept only as a citation of an existing, already-bounded program (ADD/Randall–Sundrum) and the connection it offers to Σ_0 – not as a new mechanism.

Parked or substantially withdrawn. **H2a's original cosmic-expansion engine** specifically (not H2a as now revised, kept above): as a global origin story for Earth's (or any body's) own field, the magnitude and shape objections in §4.3, sharpened in §6, are decisive as that version was stated. What remains viable from it is only the much narrower, already-standard river-model statement that space flows *locally* toward a mass – which predicts nothing beyond GR, and which curvature-as-cause (H2a's revised form) now supersedes rather than needs. **H4**'s literal claim, "a black hole at the center of every body," is directly contradicted by existing helioseismological, seismological, and geodetic data (§4.5) and is withdrawn in that form; the only reading not already ruled out is the micro-depression row of the taxonomy with no horizon, at which point H4 is ordinary gravitational instability under a new name. **H6**'s claim that its central vortex reaches the scale where flat rotation curves are measured is withdrawn as stated (a three-to-four-order-of-magnitude scale mismatch, §4.7); the target observation remains real and open.

Narrowed to a question, not a mechanism. **H3** and **H5** survive only as restatements of the genuine, open supermassive-black-hole seed problem, to be checked against the mainstream benchmarks (direct collapse, heavy seeds, super-Eddington accretion; stellar-mass seeds plus mergers) they currently compete with rather than build on – not as literal tears in, or literal collisions of, spacetime's fabric, for which no stress-energy source or falsifiable signature has yet been stated.

Motivated, but numerically failing as stated. **H10** is in a different position from everything above: it is neither kept as working content, withdrawn because data contradicts it, nor merely narrowed to an open question for lack of a number – it is kept because the *kind* of answer it proposes (spacetime's stiffness as an induced, emergent residue of the quantum vacuum, precedented by the Casimir effect and by ordinary surface tension, §4.11) is a genuinely better-motivated shape of "why" than simply assuming Σ_0, the way H1 and plain GR both still do. But its own numeric check, carried out honestly rather than left undone, fails by roughly thirty orders of magnitude when built from Standard-Model field content and scale. This paper records that failure as a result, not a reason to quietly drop the idea or to retroactively select whatever cutoff reproduces the right answer – doing the latter would be exactly the "papering over with a placeholder prediction" §6 opens by rejecting for H3–H6.

7.2 Open problems and known weaknesses

The strongest objection to each hypothesis, and its current status, collected from §2 and §4:

- **H1** – not yet falsifiable on its own (constant Σ makes no prediction beyond GR). *Open*, pending H1b.

- **H1b** – h and λ unspecified; Birkhoff's theorem means any resolution is an amendment to the field equations, not a parameter. *Open, and not yet written as the amendment it would have to be.*
- **H2-engine** (the retired cosmic-expansion/spin driver) – magnitude ($\sim$ 10 orders short for cH_0, $\sim$ 30 orders short for $H_0^2 R$) and shape (uniform driver vs. mass-centered, inverse-square field) objections are decisive as stated. *Answered, in the negative.* **H2-revised** (curvature itself as the cause, kept as this paper's defining statement; §4.3) is a different claim from H2-engine and is not touched by that negative answer – it is *not* answered in the negative, but it is also not a prediction beyond GR: it is the standard $-\nabla\Phi$ recovery, *open* only in the sense that it does not, by itself, explain why mass curves spacetime this way in the first place.
- **H3** – no stated mechanism distinguishable from direct-collapse seeding. *Open.*
- **H4** – literal claim contradicted by interior- structure data. *Answered, in the negative,* for bodies where that data exists.
- **H5** – survives the Bullet Cluster test only by positing near-frictionless space-collision, and lacks a stated stress-energy source. *Open, on both counts.*
- **H6** – scale mismatch between the vortex's reach and the observation it targets. *Answered, in the negative,* as stated; MOND's own split between rotation-curve success and cluster/CMB difficulty is the benchmark any revision should expect to face.
- **H7** – magnitude gap (32 orders of magnitude) at the present epoch, and no stated mechanism for net extraction. *Answered* on extraction (dropped); magnitude gap is expected, not a flaw, given the mechanism.
- **H8** – angular momentum and $R_{\rm core}$'s minimum both unstated. *Open,* but narrowly – both are additions to an otherwise well-grounded picture, not challenges to it.
- **H9** – n and ℓ_* unspecified, and already tightly bounded ($\lesssim$ 40–50 μm) by existing tests. *Open, within a narrow remaining window.*
- **H10** – cutoff sensitivity: Standard-Model field content and scale give $\Sigma_{\rm induced}$ roughly thirty orders of magnitude short of the measured Σ_0, and matching it requires a cutoff defined circularly from G itself. *Answered, in the negative, as a Standard-Model-only claim*; open only if a specific beyond-Standard-Model resolution (new field content, a different cutoff, a stated fix to the same hierarchy problem behind H9) is supplied and checked – which this paper has not yet done.

7.3 Relation to Pattern Arithmetic (if applicable)

[PLACEHOLDER or: Not applicable.]

8 Conclusion

This paper proposed treating space as a real, elastic, fluid medium (H1, H1b) and, from that single departure, reframed gravity as the felt consequence of spacetime curvature – the widening gap between local clock rates at neighboring points as curvature deepens – rather than as an added force (H2a), with H2b and H2c each showing why a spatial bend alone and a flat-spacetime rotational analogy cannot do that job on their own. Pressed against a further cluster of open astrophysical questions (H3–H9), and then, deeper still, against the question of why spacetime has any elasticity to speak of in the first place (H10), this departure survived some of its own numeric checks (H1's inverse-square recovery, H2c's match to the measured solar light-bending angle) and failed others outright (H2a's original cosmic-expansion engine, H4's literal claim, H6's scale claim, and now H10's own cutoff-sensitivity check, off by roughly thirty orders of magnitude), each resolved here in the direction its own numbers pointed rather than defended or argued around (§7). What remains, after that filtering, is not offered as a replacement for general relativity – H2a-revised is GR's own geodesic equation in its Newtonian limit, not content beyond it, H10 names the right kind of question about Σ_0's origin without yet answering it, and H3, H5, H7, H8, and H9 still owe the specific, falsifiable form §6 asks of them – but as a disciplined set of working hypotheses, each stated plainly enough to be checked and, where checked, either kept, narrowed, or withdrawn on its own merits. Mercury's perihelion precession has since been checked and matched (§4.3); equivalence-principle tests and gravitational- wave speed/waveform constraints are the next, specific numeric items this framework still owes (§5).

A Derivations

[PLACEHOLDER or: None yet.]

B Glossary of Notation

[PLACEHOLDER]

References

[1] [PLACEHOLDER citation]

[2] [PLACEHOLDER citation]